Text properties and children's reading pace in a multilingual storytelling app

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Abstract

Children increasingly read machine-generated stories on screens, and whether the text properties that govern reading under controlled conditions govern it there is unknown. We will analyse logs from *Let's Story*, a feature of the *Applaydu* app, in which children aged 7 and older read in ten or more languages. We will model reading time per word from lexical difficulty, grammatical load and referential continuity. Six confirmatory hypotheses concern these associations, their moderation by age band, narration's selectivity, children's difficulty ratings and accumulated in-app reading. Bayesian crossed-effects models will represent readings within texts, children and languages. Coefficients outside a region of practical equivalence would support the hypotheses, in either direction for the two that adjudicate between rival developmental accounts, and posteriors inside it would show that the laboratory effects do not transfer. Because families choose narration, texts and continued use, we will report every coefficient as a conditional association.

Introduction

A generation of children now meets a large share of its reading on a screen, and an increasing share of what appears there was written by a machine. Commercial storytelling applications generate a narrative to order from a handful of choices a family makes, read it aloud in the family's language, and record what the child did with it. The behavioural record such platforms accumulate is far larger than any laboratory corpus of children's reading, spans ten or more languages, and arises under exactly the conditions that laboratory work has to strip away. The count below sizes the evidence base that describes it. We ask whether the text properties that govern reading under controlled conditions still govern it in the wild, and whether their hold on the reader changes over the years in which decoding becomes automatic.

The laboratory answer is well established at three levels of linguistic description. At the level of the word, frequency of occurrence is among the most consistently replicated predictors of reading behaviour that psycholinguistics has produced. High-frequency words are recognised faster and fixated for less time than low-frequency words, and the effect on text reading is closely similar across 12 alphabetic languages (Kuperman et al., 2024). In children the effect is larger than in adults, reflecting less automatised lexical access (Joseph et al., 2013). Frequency norms drawn from subtitle corpora approximate everyday exposure well enough to have become the standard operationalisation (Brysbaert & New, 2009; van Heuven et al., 2014). Resources built from what children actually read, such as the Children and Young People's Books Lexicon, give a still more defensible baseline for developing readers (Korochkina et al., 2024). Related lexical properties behave similarly. Concrete and highly imageable words are processed faster than abstract ones in natural reading (Magnabosco

& Hauk, 2024) and are remembered better (Paivio, 1991), an advantage usually attributed to the availability of perceptual as well as verbal representations (Bernabeu, 2022; Kaup et al., 2025). Grounding matters most early in vocabulary development, since the meanings children acquire first are the more concrete ones (Muraki et al., 2025). The diversity of the vocabulary a text deploys is a further index of the demand it places on the reader (McCarthy & Jarvis, 2010), and vocabulary knowledge constrains comprehension in children more tightly than in skilled adults (Cain & Oakhill, 2014).

At the level of the sentence, structure imposes its own cost. Mean dependency distance, the count of embedded clauses and the ratio of subordinate to main clauses all index the demand a text places on the reader to hold a dependent in mind until its head arrives (Gibson, 1998). Long dependencies are expensive because the intervening material has to be carried, and because children's verbal working memory is still developing they are disproportionately affected by that cost relative to adults (Chang, 2020; Cui et al., 2024; Delage & Frauenfelder, 2019). Syntactic skill is a source of comprehension variance separate from vocabulary in the middle grades (Nielsen et al., 2025), and sensitivity to grammatical structure continues to underwrite comprehension well past the primary years: Marjokorpi and van Rijt (2025) found explicit grammatical understanding to predict reading comprehension in secondary-school students after socioeconomic background, writing ability and other literacy factors were controlled. The evidence at the grain this study uses is less settled. In the Multilingual Eye-movement Corpus, global measures of syntactic and lexical complexity, including dependency depth and embeddedness, were negligible predictors of adults' sentence and passage reading times once length and position were accounted for (Kuperman et al., 2025). Whether a text-level grammatical measure carries a reading-time cost for children reading whole stories

is therefore an open question, and Hypothesis 1 is written so that a practically null grammatical coefficient would replicate that adult result in children while a non-negligible one would qualify it.

At the level of the passage, what matters is whether the text states the connection between one idea and the next or leaves the reader to supply it. Sentence-to-sentence semantic similarity, overlap in referring expressions and the density of explicit connectives together index how much of that work the surface of the text has already done (Crossley et al., 2019). Texts that do more of it are generally easier to follow, and the benefit is largest for readers with least prior knowledge to draw on (McNamara et al., 1996; O'Reilly & McNamara, 2007). Automated tools now make these indices computable at scale (Crossley, 2025; Graesser et al., 2014), and multi-dimensional characterisations of difficulty built from them outperform the classical readability formulae they replaced (Crossley et al., 2023; Goldman, 2024; Martinc et al., 2021).

The systematic review that accompanies this protocol sharpens that general account in ways that shape the present design. Text difficulty does not reduce to word frequency: in 924 German fourth-graders, estimated lexical coverage predicted comprehension non-linearly, rising more steeply above roughly 56 per cent coverage (Ludewig et al., 2023). In 158 children aged between 8 years 8 months and 11 years 2 months, word frequency, pronoun density, concreteness and deep cohesion each explained unique variance in passage comprehension after decoding and oral language were taken into account, and lexical arousal explained variance beyond them (Pickren et al., 2022), which is the direct warrant for carrying affective and discourse variables alongside conventional readability measures. Effects also depend on what is being asked: across 1,229 children in grades 2 to 9, the direction and size of a text characteristic's contribution varied with question type (Aparicio et al., 2022). Feature-based prediction of item-level comprehension from sentence complexity, lexical

surprisal and rare-word use is already feasible (Nie et al., 2022), and syntactic complexity has been shown to impair comprehension accuracy and slow recall in bilingual children (Wasito Kirana et al., 2025).

Two further strands bear on the present design because they concern the reader and the reading context. The first is the child's history of reading. Longitudinal work consistently shows that this history cannot be read as a simple dose of practice. Using random-intercept cross-lagged models from ages 7 to 16, Torppa et al. (2020) found that weaker early reading predicted less subsequent leisure reading, while later within-person increases in book reading predicted gains in comprehension. A ten-year study likewise found reciprocal associations between skill and print exposure, with the skill-to-practice path most visible early and accumulated exposure becoming more predictive later (van Bergen et al., 2021). Accumulated use of an app therefore carries both practice and the stable differences in skill and motivation that produced the use in the first place, so we will estimate within-child change and between-child differences separately.

The second is narration. Read-aloud audio is often treated as a uniform benefit, and the evidence does not support that reading. A meta-analysis of 30 reading-while-listening studies (62 effects, total N = 1,945) found a small pooled comprehension advantage (g = 0.18), no reliable advantage where ordinary reading was self-paced (g = 0.06, 95 per cent CI [-0.07, 0.19]), and a larger one only where ordinary reading was paced by the experimenter (g = 0.41, 95 per cent CI [0.13, 0.70]) (Clinton-Lisell, 2023). Among students with reading difficulties, text-to-speech tools showed a moderate pooled effect (d = 0.35) that fell to d = 0.15 and included zero in within-participant designs (Wood et al., 2018). Neither synthesis settles the mechanism, which a large naturalistic corpus is well placed to address. If a spoken rendition works by supplying word identity without

print-to-sound mapping, it should remove much of the cost that lexical difficulty imposes while leaving the cost of connecting ideas largely intact. If narrated reading time is instead set by the duration of the recording, every text property should lose its purchase on time in the same proportion. Hypothesis 4 is written to separate these accounts.

The size of the relevant literature can be counted. On 26 August 2026, a Scopus search over 2018 to 2026 returned 3,701 records on children's reading comprehension, 6,592 on computational text features and 133,132 on platform log data. The pairwise intersections were 17, 28 and 45 records, and the three-way intersection was empty. The exact expressions are reproduced in the supplementary search record, the script that issues them and archives their counts is deposited with this protocol (Bernabeu, 2026c), and Figure 1 shows the sets. Over the same window, records combining children's reading with machine-generated text went from none before 2024 to two, six and four in the three years since. That small strand consists of classroom experiments: 106 Turkish fourth-graders who read stories generated from their own prompts gained in comprehension and motivation over controls (Balantekin, 2026), and a further trial reports similar gains for generated science stories (Bulut & Aktepe, 2026). These studies measure outcomes in the classroom and describe the texts qualitatively. The laboratory counterpart for adults is recent: a corpus of eye movements from 107 English-speaking adults reading machine-generated texts of six kinds, annotated with the generating models' internals (Bolliger et al., 2025).

Let's Story, a feature of the *Applaydu* app developed by Gameloft and Ferrero International, combines the three topics of that count. Families choose a setting, a hero, a companion, a plot and a theme, the five story attributes the platform's data dictionary records for each generated story. A generative model then writes a narrative around those choices, and the app renders it as an illustrated storybook

and, at the family's option, reads it aloud (Ratner et al., 2025). Every completed reading is logged, with the time taken, the language, the child's age band, the reading level the family has set and an optional judgement of how hard the story was. Comparable platforms have been studied one feature at a time. Booton et al. (2023) reviewed 11 studies and found inbuilt narration to aid comprehension and word learning, while text-linked interactive hotspots showed no evidence of benefit. Content analyses of the wider market are less encouraging: Furlong et al. (2025) found expert reviewers willing to recommend only 27.5 per cent of 309 phonics apps, with consumer star ratings a poor guide to expert quality, and Chatzopoulos et al. (2023) found educational content to be the dimension in which the largest share of 50 Google Play apps scored poorly. Log data from a research-oriented platform have been shown to track ability, with reading speed in the BookBites app correlating with standardised scores at $r = .39$ to .48 (Sortkær et al., 2021). The commercial edutainment sector, whose user base is far larger, has been studied through content analyses and small experiments rather than through its logs.

The theoretical commitments behind the hypotheses are held in a machine-checkable theory object, deposited with this protocol, that names twelve constructs, eighteen propositions each carrying the mechanism it claims, six auxiliary assumptions and six rival accounts, and records which prediction is derived from which proposition and which rival it is meant to separate. Figure 2 is generated from that object, so the figure and the hypotheses cannot drift apart on the counts, scores and derivation links the checks enforce. The object passes all twelve items of the versioned rigour checklist implemented in theoryforge, scoring 90.9 out of 100 (Supplementary Section S1) (Bernabeu, 2026d). Its causal graph is acyclic, so implications can be derived from it mechanically: that the three text properties are mutually independent given the reading level and the generation route, that narration

is unrelated to accumulated reading, and that the difficulty rating adds nothing about reading level once reading time is known. We will test these implications before interpreting the confirmatory models, against the criterion registered under Outcome-neutral quality checks, and we will report any failure.

This study is observational throughout. Children choose their stories, their reading level and whether to switch narration on, and families choose whether to keep using the app. A protocol built on such records has to say what that gains and what it costs, because platform data answer a different question from an experiment. A laboratory experiment establishes that a text property *can* move reading when an experimenter controls the text, the setting and the task. It cannot establish that the property *does* move reading when nobody controls anything. The second question is one about generalisation, and it cannot be settled in the setting that produced the original result (Nastase et al., 2020; Yarkoni, 2022). Explanation and prediction are best treated as complementary aims served by different designs (Hofman et al., 2017; Hofman et al., 2021), and the same holds for control and representativeness. Mining naturally occurring records is by now an established methodology with its own accumulated cautions (Goldstone & Lupyan, 2016; Paxton & Griffiths, 2017), and the account below draws on them.

The platform's first advantage is stimulus sampling. Reading research has known since Clark (1973) that treating a handful of experimenter-written texts as a fixed set inflates confidence in conclusions meant to hold for texts in general, and the mixed-model solution has been available for over a decade (Judd et al., 2012). Yarkoni's (2022) survey of the problem found that psychologists routinely model variation between participants while leaving stimuli, tasks and sites unmodelled, which constrains generalisation far more tightly than most reports acknowledge. The design here treats Text as a

crossed random factor over hundreds of texts that children selected for themselves, which is both the prescribed remedy and, in Brunswik's sense, a representative design: the stimuli are sampled from the environment the reader inhabits, where a systematic design would construct them to instantiate a factorial contrast (Dhami et al., 2004). We avoid the phrase *ecological validity* throughout, since it has been shown to carry several incompatible meanings in current use (Holleman et al., 2020; Kihlstrom, 2021). The claim made here is the narrower and checkable one about how the stimuli were sampled. The precision analysis makes the same point in reverse, since every text coefficient is bounded by how many distinct texts there are, however many sessions the extract holds.

Linguistic breadth is the second gain. English dominates cognitive science to a degree that warps which mechanisms are studied and licenses overgeneralisation from one atypical language to the species (Blasi et al., 2022). Reading is among the worst affected, because writing systems differ in ways that matter for every stage of processing. The most ambitious response to date, the Multilingual Eye-movement Corpus, assembled eye-tracking during text reading in 13 languages in its first wave through an international multi-laboratory collaboration, and its authors open by reporting that the existing empirical base is insufficient for cross-linguistic comparison (Siegelman et al., 2022). A second wave has since extended it to 21 languages, adding logographic and abugida writing systems (Siegelman et al., 2025). A commercial platform reaches many of the same languages incidentally, in far larger numbers of readers per language, and reaches children where the corpus records skilled adults. The comparison must be stated precisely, because MECO records eye movements and this study records only whole-story duration. The platform gains breadth and loses resolution, and a given question needs one or the other.

The third gain is behaviour as it occurred. Where the alternative measure of how much or how fast a

child reads is an adult's recollection, the log is the better instrument: logged and self-reported digital media use diverge systematically enough to have warranted a meta-analysis of the discrepancy (Parry et al., 2021), and in 109 French primary-school children parents' estimates of weekly reading time were three times the logged figure, which predicted fluency better (Brossette et al., 2026).

Against this stands the strongest statistical objection to any large non-random dataset. Meng's (2018) decomposition shows that the error of a sample mean is the product of three quantities. One indexes data quality, as the correlation between the recorded value and whether it was recorded at all. The other two index data quantity and the difficulty of the problem. Probability sampling holds the first near zero. Once it is not held there, the population size stops cancelling: for any given sample size, estimation error then grows with the square root of the population. Bradley and colleagues (2021) gave the canonical demonstration: two surveys collecting roughly 250,000 responses a week and 75,000 a fortnight overestimated United States COVID-19 vaccine uptake by 17 and 14 percentage points, with margins of error so small that the wrong answers looked precise, while a well-designed panel of about 1,000 responses a week was accurate. Scale converts a small bias into a confident error, and it did so at Google before it did so in the pandemic (Lazer et al., 2014). Every selection gate in this study is a candidate for that correlation: who owns a tablet, who installs an app tied to a confectionery brand, who registers consent for research use, who switches narration off, and who finishes a story.

That objection is real, and it also has a boundary, which must be drawn accurately. Meng's result concerns the estimation of a population mean, and this study estimates no population mean. Its estimands are conditional associations inside the sample, such as the change in log reading time per word for a one-standard-deviation change in lexical difficulty, within a language and within a

reading level. A sample that is unrepresentative in who it contains biases such a slope only when selection depends on the outcome given the predictors, which is a narrower condition than the one that wrecks a mean. The consequence falls on the claim: we will report these coefficients as estimates for children who read on this platform, and we will not present them as population values for children in general, nor will we report a population mean of reading speed for any group.

One selection gate does meet the narrower condition, and it deserves separate treatment. The extract contains completed readings only, so the analysis conditions on completion, which is plausibly a common effect of the text's difficulty and the reader's capacity. Conditioning on a common effect can induce an association between its causes where none existed (Rohrer, 2018). The direction here is predictable: if harder texts are abandoned more often, the hard texts that survive into the data are disproportionately those read by children who could manage them, which attenuates the very coefficients Hypothesis 1 predicts. The bias therefore works against our own prediction, so a non-negligible result survives the objection more comfortably than a null does, and the practical consequence is that an inside-region verdict for Hypotheses 1 to 3 must be reported with the completion rate beside it. That rate is question (k) to the data controller.

Nothing here is randomised, and we do not use causal verbs for variables that families chose. That restraint is not the same as pretending the questions are not causal: euphemism has been argued to damage observational research by obscuring what is actually being asked (Grosz et al., 2020; Hernán, 2018). We state the causal structure explicitly in a deposited graph, derive the adjustment set from it, register the rival explanations it implies, and then report associations conditional on that adjustment. Estimating the text coefficients within reading level, for instance, is a consequence of the graph, and the graph itself entails it.

Secondary data also invite the analytic flexibility that preregistration exists to remove, and the field has developed specific standards for preregistering analyses of data that already exist (Van den Akker et al., 2021; Weston et al., 2019). The arrangement described under Data provenance goes further on the point that matters most. The extract is unanalysed because it is absent. The data controller holds it, and it reaches us only after in-principle acceptance, so the most common objection to observational work, that the analyst saw the data before choosing the analysis, does not arise.

When 160 teams applied machine learning to a rich birth-cohort dataset to predict six life outcomes, the best predictions were barely better than a simple benchmark (Salganik et al., 2020). This study accordingly makes a narrow claim. It does not attempt to predict a child's reading, but to estimate a handful of pre-specified associations whose attainable precision we calculated before the data were released, and to say in advance which of them the available records cannot settle. A last limit applies to any aggregate: group-level associations need not describe any individual, and variance around an expected value has been found to be two to four times larger within people than within groups (Fisher et al., 2018). This is why tier 2 separates each child's own change from the differences between children, and why the two are never reported as a single quantity.

Three consequences of all this are built into the design. First, the reading level a family sets determines which version of a text the platform serves, so text difficulty is partly assigned by a choice that also reflects the child's capacity, and we will therefore estimate the text coefficients within reading level and check whether they survive that restriction. Second, the app records elapsed time for a whole story and never for a page, so the outcome is a rate, and we state below which analyses that forecloses. Third, the data carry no measure of any individual child's reading skill,

vocabulary or language background, so age band stands in for a bundle of developing capacities and is interpreted as nothing finer.

Six confirmatory hypotheses follow, set out with their sampling and analysis plans, the precision each test requires and the reading of every possible outcome in Table 1. Two of them, the interactions between the text properties and the age band, are registered as two-sided tests that adjudicate between rival accounts, because the direct developmental evidence in this age range supports neither sign. Three text components carry the confirmatory predictions. LexicalDifficulty and GrammaticalLoad are oriented so that higher scores mean a more demanding text, and ReferentialContinuity so that higher scores mean a text that states more of the links between its ideas. Signs are fixed before any component score is joined to an outcome. The age band and narration enter the models as deviations from their observed prevalence: a reading scores 1 for the older band, or for narration on, and 0 otherwise, minus the proportion of tier-1 readings in that state. Each component's main effect is then its prevalence-weighted average slope across the two states, which carries no design effect from imbalance, and each interaction is the difference in slope between the two states, positive when the slope is larger in the older band or under narration. The outcome for Hypotheses 1 to 4 and 6 is log reading time per word, standardised within language, so a coefficient is a within-language standardised slope and the region of practical equivalence is 0.05 standard deviations. Hypothesis 4 concerns a difference between two attenuation proportions, with a region of 0.05 on that difference. Hypothesis 5 uses the child's own three-point difficulty rating on an ordinal scale, where the region is a half-width of log(1.05) on the log-odds scale, an odds ratio between 0.952 and 1.05.

Because the two main effects run in opposite directions, the same developmental claim takes opposite signs in the two interaction terms. Grammatical load is associated with slower reading, so a weaker

association in older children makes a positive slope smaller and the interaction negative. Referential continuity is associated with faster reading, so a weaker association in older children makes a negative slope less negative and the interaction positive. Wherever an account is described below as predicting that a text property bears less on the older reader, the sign of the corresponding coefficient follows from this convention alone.

The outcome is the logarithm of reading time, for a reason the hypotheses depend on. Children in the two age bands differ in overall reading rate, and a difference in rate that is purely multiplicative manufactures a spurious interaction on a raw time scale (Faust et al., 1999). On the log scale a multiplicative difference in rate appears as a main effect and never as an interaction, so every interaction reported here is a difference in *proportional* cost. A second feature of the outcome is fixed for a related reason. In the standard cohesion manipulation the more cohesive version of a text is 20 to 40 per cent longer, so a duration aggregated over a whole text would show higher coherence slowing reading for arithmetic reasons alone. Dividing by word count and retaining log text length as a covariate closes that path, and a raw duration is never used as an outcome.

Table 1 sets out each hypothesis against the sampling and analysis plan that tests it, the precision the design analysis says that test requires, how each possible outcome will be read, and what would be shown wrong. We fixed every row before any outcome data existed.

Table 1

Study design table. Eligibility for every row: sessions the platform flags as incomplete, crashed or duplicated, and accounts flagged as duplicates under the platform's own rules, are removed; a story needs a positive word count; a reading faster than 400 or slower than 10 words per minute is excluded; only an account's first reading of a text enters tier 1; and readings sharing a session-timeout value are treated as right-censored where the registered rule triggers. No datum is ever replaced.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Are the text properties that govern reading in the laboratory associated with reading pace in an app?	H1. LexicalDifficulty and GrammaticalLoad are positively, and ReferentialContinuity negatively, associated with log reading time per word.	Tier 1: every eligible completed reading in the frozen extract, one row per child and text, children recorded in age band 7-8 or 9+, under the eligibility rules in the table note. No stopping rule; the extract is fixed before modelling. ROPE of 0.05 *SD*. A determinate verdict on an effect of 0.10 *SD* with probability 0.90 needs 738 distinct texts for the lexical coefficient (95 per cent interval 640 to 843), 660 for grammatical load (593 to 731) and 634 for continuity (568 to 704); at 0.95, 926, 812 and 784. A threshold will be re-derived if the review's pooled estimate for that component falls below 0.10 *SD*, as registered under Design and precision analysis. Below its threshold a coefficient is declared underpowered before fitting.	The tier-1 Bayesian model specified under Confirmatory models: within-language standardised log time per word on the three components, the age band, narration and their interactions, adjusted for reading level, text length, valence, arousal, narrativity and provenance, with varying intercepts for Text, Language, Country and App version, varying component slopes by Language, and an Account intercept where the estimability rule is met. The three component posteriors are the registered tests.	Each coefficient is labelled non-negligible in the predicted direction, non-negligible in the opposite direction, practically equivalent to zero, or inconclusive. A combined H1 verdict is recorded only when all three are non-negligible in the predicted direction. A coefficient non-negligible in the opposite direction counts against the corresponding proposition of the theory object, and for continuity is read as a whole-text reverse cohesion effect. Equivalence is evidence that the property's laboratory effect does not reach 0.05 *SD* in this setting. Shown wrong if all three are equivalent: that the lexical, grammatical and discourse determinants of reading established under laboratory control operate on unsupervised reading of machine-generated stories.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Is the association between grammatical load and pace proportionally smaller, or larger, in the older band?	H2. GrammaticalLoad × AgeBand, tested two-sided. The capacity account predicts a negative interaction, the experience account a positive one.	As H1. Tier 1 admits only readings with a recorded age band. A determinate verdict on an interaction of 0.10 *SD* needs 1,230 distinct texts (1,107 to 1,359) at twelve readings each for 0.90 and 1,510 (1,349 to 1,680) for 0.95; declaring equivalence needs about 1,600. Both figures assume a balanced band split and will be multiplied by $1/(4p(1-p))$ at the observed band prevalence before fitting.	The same tier-1 model. The GrammaticalLoad × AgeBand posterior is the registered test. Cross-validated comparison against a model without the interaction will be reported alongside and does not gate the coefficient.	Negative and outside the ROPE supports the capacity account, and positive and outside supports the experience account. Inside the ROPE is evidence that neither developmental signature reaches 0.05 *SD* across these two bands, with the caveat that a non-monotonic trajectory cancelling across two coarse bands would produce the same posterior, which the ordinal-age sensitivity analysis addresses where the field permits. Shown wrong: whichever account the sign contradicts; a posterior inside the ROPE weighs against both, since each predicts a signed proportional change.
Is explicit connection between ideas associated with a proportionally larger gain in pace for the younger or the older reader?	H3. ReferentialContinuity × AgeBand, tested two-sided. The knowledge account predicts a positive interaction, the cue-utilisation account a negative one.	As H2. A determinate verdict needs 1,431 distinct texts (1,276 to 1,596) for 0.90; the 0.95 target was not reached within the 1,600 texts simulated. The probability of correctly declaring equivalence approaches 0.90 at 1,600 texts. The band-prevalence adjustment applies as for H2.	The same tier-1 model. The ReferentialContinuity × AgeBand posterior is the registered test. Connectives enter the component by relation type, with causal and contrastive connectives as anchors.	Positive and outside the ROPE supports the knowledge account, on which continuity substitutes for an inference the younger reader finds costly. Negative and outside supports the cue-utilisation account, on which a connective must first be reached and understood before it can help. Inside the ROPE is evidence for neither, with the same non-monotonicity caveat as H2. Shown wrong: whichever account the sign contradicts.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Is narration associated with a proportionally smaller lexical coefficient than the continuity coefficient?	H4. The proportional attenuation of the lexical coefficient under narration exceeds that of the continuity coefficient by more than 0.20.	As H1, evaluated only where both narration states occur within the eligible sample. The within-account contrast becomes the primary estimate when the accounts observed in both states cover at least the adjusted text requirement. ROPE of 0.05 on the difference of proportions. The simulated figure of 1,234 texts (1,113 to 1,361; 1,506 at 0.95) prices the lexical-by-narration interaction coefficient and is a lower bound for the difference rule; it will be multiplied by $1/(4p(1-p))$ at the observed narration prevalence, about 3.0 at an unnarrated share of 0.09, and stated before fitting.	Draw-by-draw contrast of the two proportional attenuations, computed from the derived narrated and unnarrated slopes of the tier-1 model under the rule given under Confirmatory models. Gated on the derived unnarrated slopes of the lexical and continuity components both being practically non-negligible. Reported with the residual standard-deviation ratio and repeated in the narrated sessions the audio cannot have paced.	At least 95 per cent of the posterior of the difference above 0.20 supports selective removal of decoding cost. More than 95 per cent inside the region supports the account on which the recording sets the pace. More than 95 per cent below -0.05 is reverse selectivity, contradicts both accounts and is reported as such. Anything else is inconclusive and both accounts remain live. If the gate fails, H4 is reported as not evaluable. If the compression diagnostic fires, the full-sample verdict is withheld and H4 rests on the unpaced subset. Shown wrong: either the decoding-scaffolding account of narration or the audio-pacing account; the contrast is constructed so that they cannot both survive.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Do children's own difficulty judgements track the same text properties?	H5. LexicalDifficulty and GrammaticalLoad are associated with higher odds, and ReferentialContinuity with lower odds, of a harder rating, each outside the region, and the three rank by magnitude as in the reading-time model.	Tier-1 sessions carrying a rating. A text rated by fewer than five distinct children does not contribute. The primary model is complete-case, with an inverse-probability sensitivity analysis for non-response. ROPE of a half-width of log(1.05) on the log-odds scale. At the 30 per cent availability the structural sample showed, the 0.90 target is not reached within the 1,600 texts simulated and projects to roughly 8,000 to 8,700 generated texts, about 2,300 to 2,500 carrying at least five ratings (Supplementary Section S5). The simulation will be re-run at the observed availability before fitting. This is the hypothesis most likely to be declared underpowered.	Ordinal cumulative-logit model with the same fixed and grouping structure and language-varying thresholds. The ordering claim is the posterior probability that the three coefficients rank by magnitude as they do for reading time, read against the 0.90 criterion registered under Decision rules.	All three odds ratios outside the region in the predicted directions and ordering probability at least 0.90: two independent measurements converge on the same ordering of text demands. Odds ratios in the predicted directions but ordering probability below 0.90: partial convergence; the rating is usable as a difficulty signal but not as a rank measure. Odds ratios inside the region while the reading-time coefficients are non-negligible: the rating is not a usable measure of experienced difficulty, and the ordering is not interpreted. Both outcomes null: the components themselves are suspect. If underpowered, the ordering is reported descriptively and convergent validity is recorded as not established. Shown wrong: that reading time and reported difficulty are both readouts of the same expended effort.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Within a child, is more accumulated in-app reading associated with a faster pace?	H6. Within a child, accumulated in-app reading is negatively associated with log time per word.	Tier 2: children with at least three eligible sessions on distinct texts. The simulated requirement is 563 children (463 to 674) at five readings each for 0.90 and 681 (543 to 834) for 0.95, for the unadjusted trend; the tier-2 sweep will be re-run at the observed sessions-per-child distribution with the registered covariates added, and the hypothesis is declared underpowered, not untestable, below the re-run threshold.	The tier-1 specification refitted on tier 2 with a child varying intercept, decomposing accumulated words into the child's mean and the session's deviation from it, alongside session order, days since the previous session and days since the child's first eligible session. The deviation term is the registered test.	Negative and outside the ROPE: an association over the observation window, never a practice effect, because skill and practice are reciprocal over longer spans and maturation is the named residual alternative. Positive and outside: contradicts the consolidation account and is read as within-window slowing that the covariates do not capture. Inside the ROPE: evidence against a within-window consolidation signal of at least 0.05 *SD*. Shown wrong: that reading consolidates the recognition machinery reading depends on, within a window short enough for the reverse path to be unlikely to dominate.

Hypothesis 1: text properties and reading pace. We predict that, across distinct texts and within reading level and language, LexicalDifficulty and GrammaticalLoad will each show a positive association with log reading time per word, and ReferentialContinuity a negative one. The three coefficients are evaluated and labelled separately, and the failure of one does not suppress reporting of the others. An overall verdict on Hypothesis 1 is recorded only when all three are practically non-negligible in the predicted direction. A coefficient that is practically non-negligible in the opposite direction is read as evidence against the corresponding proposition of the theory object, and a positive continuity coefficient in particular as a whole-text reverse cohesion effect of the kind reported for high-knowledge readers (McNamara et al., 1996; O'Reilly & McNamara, 2007). Such a coefficient blocks the combined verdict without suppressing the others. A coefficient practically equivalent to zero is evidence that the property's laboratory effect does not reach 0.05 standard deviations in this setting, which for the grammatical component would extend to children the negligible passage-level effect of global complexity measures reported for adults (Kuperman et al., 2025).

Hypothesis 2: grammatical load and the age band. We register the GrammaticalLoad × AgeBand interaction as a two-sided adjudicating test, because two accounts of what changes between these ages predict opposite signs on the scale this study uses. On the capacity account, a grammatical dependency costs time in proportion to how close its maintenance comes to exhausting the working memory available, so as that capacity matures the same structure costs proportionally less and the interaction is negative (Delage & Frauenfelder, 2019; Peng et al., 2018). On the experience account, what changes is exposure: an older reader has met complex structure far more often and has been taught the punctuation that marks its boundaries, engages with it more fully in consequence, and

pays a proportionally larger cost, which makes the interaction positive (MacDonald & Christiansen, 2002; Montag & MacDonald, 2015).

The direct evidence does not decide between the accounts. The largest eye-movement sample over this band, 81 children reading 12 items, found no attenuation at all, but it manipulated temporary ambiguity, which indexes reanalysis and not the sustained maintenance that dependency length imposes, so its null does not transfer to the present operationalisation (Wonnacott et al., 2016). Where an interaction has been reported it was confined to late measures with fourteen children per group (Joseph & Liversedge, 2013). That study's tables let the effect be expressed as a proportion of baseline, and the two measures disagree. On go-past time in the target region, our computation gives 15 per cent in the younger children, 25 per cent in the older and 31 per cent in adults, which is the pattern a positive interaction would reproduce on a proportional scale. On post-target total reading time, where the study's own significant interaction lay, the proportional effect ran the other way, from 27 per cent in the younger children to 12 per cent in the older and 4.5 per cent in adults. A third possibility is that development over this band is not monotonic at all: the largest of the existing tests, with 250 children aged 8 to 11, found the processing of complex sentences to change unevenly across that range (Booth et al., 2000). A negative interaction outside the region of practical equivalence supports the capacity account, a positive one supports the experience account, and a posterior lying inside the region is evidence that neither developmental signature reaches a detectable size across these two bands. That third outcome carries a caveat: a non-monotonic trajectory cancelling across two coarse bands would produce the same posterior, which is why the ordinal-age sensitivity analysis is registered under Sensitivity analyses. It would still be informative, since the existing tests of this interaction range from 14 children per group to 250, and the eye-movement tests to 81.

Hypothesis 3: referential continuity and the age band. We register the ReferentialContinuity × AgeBand interaction the same way, because the construction-integration framework supplies two moderators of coherence facilitation that both rise between 7 and 11 and push in opposite directions. On the knowledge account, a reader who cannot bridge a gap from what they already know depends on the text to state the connection, so surface continuity is worth most to the younger reader and the interaction is positive, the facilitation being a negative slope that becomes less negative with age. On the cue-utilisation account, exploiting a connective requires enough decoding fluency to reach it and enough explicit knowledge of what it signals, so the older reader gains more and the interaction is negative. The clearest evidence that strategic, high-knowledge readers gain from cohesion comes from undergraduates (O'Reilly & McNamara, 2007).

Here too the existing evidence does not decide between the accounts. Eight- and ten-year-olds gained the same speed-up from an appropriate connective in self-paced reading, with the age interaction explicitly null across 91 children (Cain & Nash, 2011). Online sensitivity to a coherence break did not differ between children of 8 to 9 and children of 10 to 11 (Helder et al., 2016). In the demonstration of a reverse cohesion effect in fourth-graders, the effect of cohesion turned on world knowledge and depended far less on decoding skill, with age untestable within a single grade (McNamara et al., 2011). Reading rate under a whole-text coherence manipulation has, to our knowledge, not been measured in this age range. The three outcomes are read as for Hypothesis 2. Because the sign of a coherence effect also depends on which cohesive device carries it, connectives enter by relation type and the component's anchors are the causal and contrastive connectives, for the reason given under the dimension reduction.

Hypothesis 4: the selectivity of narration. We predict that the LexicalDifficulty coefficient will

be proportionally smaller under narration than the ReferentialContinuity coefficient is. Writing π_{lex} and π_{cont} for the proportional attenuation of each coefficient under narration, computed draw by draw from the posterior under the rule given under Confirmatory models, the prediction is that $\pi_{\text{lex}} - \pi_{\text{cont}} > 0.20$. A spoken rendition supplies word identity without print-to-sound mapping, which is the step that a low-frequency or long word makes expensive, and it does not supply the link between two ideas, which the reader must still construct. Reading aloud the directions, proper nouns and questions of a test while the child still decodes the passages helps poor decoders with adequate listening comprehension, and pacing alone did not produce the gain (Giusto & Ehri, 2019). The benefit of text-to-speech is largest for children whose decoding is weak but whose language is intact (Keelor et al., 2023), and within a narrated passage the time saved falls on the first encounter with an orthographically novel word and has gone by the second (Valentini et al., 2024).

The competing account holds that narrated reading time is set by the duration of the recording, in which case every text coefficient shrinks by the same proportion and $\pi_{\text{lex}} - \pi_{\text{cont}} = 0$. That account has substantial support of its own. The aggregate comprehension benefit of reading while listening is confined to studies whose comparison condition was externally paced (Clinton-Lisell, 2022; Clinton-Lisell, 2023). A preregistered test of the resource-freeing mechanism in adult second-language readers found comprehension *worse* under narration than in silent reading (Hui & Godfroid, 2026). Where the compression has been measured incidentally it is severe: the standard deviation of exposure time fell from 11.56 seconds when self-paced to 0.62 seconds when narrated (Valentini et al., 2024). Neither account has been tested on the contrast that would separate them, which is reading time regressed on decoding-level and on integration-level text predictors separately, in narrated and unnarrated reading of the same texts.

Four outcomes are registered, with the posterior criteria given under Decision rules. A difference above 0.20 supports selective removal of decoding cost. A difference practically equivalent to zero supports the account on which the recording sets the pace. A difference determinately below -0.05, meaning that narration removes more of the continuity cost than of the lexical cost, is reverse selectivity, which contradicts both accounts and will be reported as such. Anything else is inconclusive and both accounts remain live. The threshold of 0.20 is four times the region's half-width and is a registered convention.

Who reads without narration is itself a selection. Narration was on in about nine sessions in ten of the structural sample, so the unnarrated slopes, which are the denominators of both attenuation proportions, are estimated largely from the families who chose to switch the audio off. If those are disproportionately the fluent readers, their slopes are attenuated by selection and both proportions are distorted, and the data carry no measure of reading skill with which to adjust for this. We will therefore report narration prevalence by age band, reading level and language before fitting, and the rule under Confirmatory models promotes the within-account contrast to the primary estimate where the data allow it. Two further guards, a registered compression diagnostic and a repeat of the contrast in the narrated sessions the audio cannot have paced, are specified there as well. Because the contrast is a ratio, it is evaluated only when the unnarrated slopes of the lexical and continuity components are both practically non-negligible. If either is not, there is no attenuation to compare and Hypothesis 4 is reported as not evaluable. This gating rule does not depend on the sign of any coefficient.

Hypothesis 5: the child's own judgement of difficulty. We predict that, in an ordinal model of the post-reading rating, LexicalDifficulty and GrammaticalLoad will be associated with higher odds of

a harder rating and ReferentialContinuity with lower odds, each by an odds ratio outside the region, and that the three standardised coefficients will rank by magnitude as they do in the reading-time model. The second clause makes this more than Hypothesis 1 repeated on a second outcome. Two measures of the same three constructs, one behavioural and one introspective, obtained on different scales from overlapping sets of children, should rank those constructs the same way, and we will report the posterior probability that they do as the study's principal convergent-validity check. That probability is read against a registered criterion: at least 0.90 counts as agreement on order, and anything lower counts as order not established.

Two facts about the rating constrain the design, and neither is established in the literature. To our knowledge, no published study has had children of this age give a global post-reading judgement of how difficult a text was and validated it against the text's measured properties: the child metacognition literature rates adjacent things, such as confidence in one's own answers, judgements of learning for word lists and enjoyment of an activity. The three-point easy, just right and difficult format is a classroom heuristic with no published psychometric validation, and nothing licenses assuming that a single rating from a single child is reliable. That children can validly judge text difficulty is therefore something this study tests, and not something it assumes.

For the design, the effective sample size is the number of distinct children rating each text, and the raw count of ratings overstates it, so the model carries a text varying intercept, and we will report the distribution of raters per text before fitting, applying the five-rater floor registered under Confirmatory models. We will report the intraclass correlation for the rating at text level, a reliability estimate for this format.

For interpretation, a failure here is ambiguous on its own, and Hypothesis 1 resolves it. If the

text components predict reading time but their odds ratios for the rating lie inside the region, the feature pipeline is sound and the rating is not a usable measure of experienced difficulty, which is worth reporting given how widely such single-item checks are deployed in educational software, and the ordering is then not interpreted. If the odds ratios lie outside the region in the predicted directions but the ordering probability falls below 0.90, the convergence is partial, and the rating is usable as a difficulty signal though not as a rank measure. If the components predict neither outcome, the components themselves are suspect. If they predict both and in the same order, two independent measurements converge on the same ordering of text demands. All four readings are recorded here in advance. This hypothesis is also the one the design analysis marks as most likely to be underpowered. An inconclusive verdict at the census is itself informative, because it bounds the reliability a single-item difficulty check can have at the availability such apps achieve. In that case the ordering is reported descriptively and convergent validity is recorded as not established, so that the fifth outcome-neutral check has a defined outcome whether or not Hypothesis 5 reaches a verdict.

Hypothesis 6: accumulated reading and pace. We predict that, among children with several recorded sessions, a doubling of a child's own accumulated in-app reading will be followed by a reduction in that child's log reading time per word, over and above session order, the interval since the previous session and the time elapsed since the child's first session. On the consolidation account, every encounter with a written word form sharpens the representation supporting its later recognition, so reading consolidates the machinery that makes reading cheap. The relation between practice and skill is reciprocal over years (Torppa et al., 2020; van Bergen et al., 2021), so the coefficient is an association within the observation window, and never an estimate of a practice effect. Ordinary

maturation over a months-long window would also make a child faster, and accumulated words rise with elapsed time within every child. The elapsed-time covariate is what separates the two, and maturation remains the named residual alternative wherever the coefficient is reported. A positive within-child coefficient outside the region contradicts the consolidation account and is read as within-window slowing that the covariates do not capture. A posterior inside the region is evidence against a within-window consolidation signal of at least 0.05 standard deviations. This hypothesis depends on a subsample whose size is not known before the extract is released, and it is declared underpowered, not untestable, if that subsample falls below the threshold re-derived for it under Design and precision analysis.

Every hypothesis is labelled under the single posterior rule given under Decision rules and diagnostics, and an inconclusive result is reported as such. No hypothesis is filtered by a preliminary significance test. Three labels apply before a verdict. A hypothesis is declared untestable, before its model is fitted and without reference to any coefficient, if its component fails the construction rule set out under the dimension reduction or if its outcome or contrast is absent from the frozen extract. A hypothesis whose data fall below the precision threshold recorded for it in Table 1, as re-derived under Design and precision analysis where that section says so, is instead declared underpowered: its model is still fitted and reported, and no conclusion is drawn from an interval the design analysis predicted would be indeterminate. A hypothesis is reported as not evaluable when a registered post-fitting contingency removes its test, which happens in two cases only: the Hypothesis 4 gate, and failure of the positive control on the timing column.

The limits of what follows should be equally clear. This study will not identify a causal effect of any text property, age band or narration setting. It will not estimate how fast children in general read,

nor generalise beyond families who chose to install and use this application. It will say nothing about comprehension, since none of the contingent hypotheses is activated at submission. Where the data controller’s answers falsify an auxiliary assumption, the affected predictions are reassigned by the rules in Table 2, never reinterpreted after the fact.

Methods

Ethics information

This research complies with all relevant ethical regulations. Data are generated in the ordinary operation of a commercial application and are collected by its publisher under the terms of use and privacy policy a family accepts on installation, with consent for research use of anonymised records registered in the app by a parent or guardian. That registration is the informed consent under which the records reach us, and no further consent is sought because no participant is contacted. We play no part in collection, have no contact of any kind with any child or family, and return nothing to the platform, so nothing done here can alter what any child experiences. No compensation is provided or applicable, because the records arise from families' ordinary use of the app. The extract supplied to us carries no direct identifier: the account key is a pseudonymous string generated by the platform, and no name, contact detail, precise location or device identifier is included. Ethical approval for secondary analysis of these anonymised records [has been granted by / is being sought from] the Departmental Research Ethics Committee of the Department of Education, University of Oxford (reference [to be inserted]), and will be in place before acceptance in principle. The study involves no experimental manipulation and no intervention, so it raises no question of withdrawal from a procedure. A family's right to have its records erased is exercised through the platform, and any erasure occurring between the building of the extract and its transfer will be recorded by the data controller and reported at Stage 2.

Two conventions hold throughout this section. First, no decision about whether a hypothesis is tested, on which sample it is tested, or with which model, depends on a fitted coefficient: every such

decision is a function of provider-computed counts, of written answers from the data controller, or of thresholds registered here, with one exception, the Hypothesis 4 gate. Four registered contingencies read fitted quantities after fitting and are listed under Decision rules and diagnostics. Second, wherever the text says a quantity will be reported, we will report it in the Stage 2 manuscript whether or not it changes a conclusion.

Pilot data

The pilot material is a 35-row structural sample of the session table, the data dictionary, and a summary of that sample computed with the count script described under Data provenance. The first two were received from the data controller on 27 July 2026 to confirm the shape of the table, and the third we computed from the sample on 6 August 2026. The sample carries real reading times for 35 sessions on five distinct texts, one session per account, in ten reading languages. It cannot support any hypothesis in this protocol: none is identified by 35 observations, and every within-child term requires accounts with more than one session, of which it contains none. It fixed three inputs of the design analysis, which are the narration prevalence (on in 32 of 35 sessions), the rating availability (present in 10 of 35) and the age-band missingness (blank in 10 of 35), and Supplementary Section S8 reports it in full. The sample is deposited for peer review under embargo until Stage 2 acceptance, at the location the Data availability statement names.

Design

Data provenance, sequestration and outcome blindness

The data this protocol analyses already exist. Nature Human Behaviour admits secondary analyses of existing data on condition that authors can evidence no prior access to the data in question. We

put the arrangement set out below to the journal's editorial office, which confirmed it as appropriate on 3 August 2026, before we wrote this protocol.

The data controller builds the export now, on its own release cycle, and holds it. Nothing of the session-level file reaches us until the journal confirms in-principle acceptance in writing. In the interval, three things may pass to us, each of which the editorial office agreed may be received in advance. The first is a manifest: the column list, the row count, the cut-off timestamp, the build date and a SHA-256 checksum for every file, so that the file eventually transferred can be shown to be the file that was built. The second is the story texts, keyed by story identifier, reading language, reading level and app version, with page breaks preserved, because every linguistic measure in this protocol is computed on them and none of them contains a behavioural outcome. The third is a set of aggregate counts computed by the data controller from the export by running a script we supplied, which reads nineteen categorical columns and the reading-time column, writes one small table of counts and a page drawn from it, and never reports a reading time as a value. At submission we hold the story texts, received on [date]. The manifest and the provider-computed counts [had been received on DATE / had not yet been received], and the count of distinct texts with eligible readings in the extract, which only those counts settle, is reported in Table 2 as unresolved: the texts we hold are keyed by identifier, language, reading level and app version, but which of them carry eligible readings, and how many readings each carries, is a fact about the extract and not about the texts.

The counts are described in detail because a count computed from an outcome column is where blindness would fail. The script treats reading time only as a column whose usability has to be established. It reports how many values are absent, non-numeric, negative or zero, and how many fall into each of seven bands whose boundaries are successive powers of ten, together with the

number of distinct values the column holds and the largest number of sessions sharing any single value. It never crosses reading time with language, age, text, account or date. Bands ten times wider than one another cannot distinguish a fast text from a slow one. What they can distinguish is a column recorded in seconds from one recorded in milliseconds, which the analysis has to know before it can be specified at all. The count of sessions sharing one value is how a session-timeout cap makes itself visible, and the sampling plan below turns that count into a registered decision about censoring.

The structural sample and the summary computed on it, described under Pilot data, are part of the record. At the editorial office's specific direction, we have not run the dimension reduction specified below and will not run it before submission, although we hold the texts on which it would operate. The component solution is therefore fully specified here and entirely unfitted.

The data controller will supply a signed certification naming an owner and a deputy for the held file, and stating its build date, its window, its row count and its checksums. The same document certifies three things about what has happened to the file. It certifies that the file has not been transferred in whole or in part to the research team or to anyone acting for it. It certifies that the file will be transferred only on written confirmation of in-principle acceptance. And it certifies that no analysis relating story text to reading time has been conducted by the controller or reported to the team. It is reproduced in the supplementary material.

Materials

The reading materials are the story texts themselves. A family chooses a setting, a hero, a companion, a plot and a theme, and a generative model writes a narrative around that combination. The text is then fixed, and it is not rewritten when the same combination is chosen again. Texts also exist

in a fixed classic set that families may read instead of a generated story. Alongside the texts, the extract records which of the two a reading drew on, the reading level the platform was serving, the language, the app version and the page count. Every session will be joined to the features of the exact version it was served, matched on story identifier, reading language, reading level and app version, so no model uses features computed on a version of a text the child did not see.

For identification, a text is a stimulus shared by many readers, and every text coefficient in this protocol is identified by comparing texts. How many distinct texts carry eligible readings is therefore the single quantity that governs how much this design can establish, and it is not yet known. The data controller's own estimate spans a few hundred to a few million, according to whether one combination of chosen attributes maps to one text shared by everyone who picks it or to a separate text per session. The published description of the platform before launch is the prior on both questions: a catalogue of over 40,000 English-language stories generated from the element combinations, many of them translated into the child's home language (Ratner et al., 2025). If that description holds for the window, texts are shared across sessions and differ across languages by translation, which would trigger the varying intercept for the underlying story crossed with Language that Table 2 registers under question (d). The controller's written answer supersedes it. The design analysis below is built around that uncertainty, and Table 2 records what follows from each answer.

For every text, we will compute features in four families. *Lexical* features are mean log frequency of content words, the proportion of words with a frequency below one per million tokens on the uniform frequency resource, mean rated age of acquisition, mean orthographic length, mean per-token surprisal under a language-specific neural language model, lexical density, the Measure of Textual Lexical Diversity and concreteness. *Grammatical* features are mean dependency distance

from a universal-dependencies parse, clauses per sentence, the ratio of subordinate to main clauses, mean sentence length and mean depth of noun-phrase modification. *Discourse* features are mean semantic similarity between adjacent sentences from a multilingual sentence encoder, the proportion of adjacent sentence pairs sharing a referring expression, connectives per hundred words by relation type and narrativity. *Affective* features are the mean and the variability of valence and of arousal.

Supplementary Table 3 gives the primary and the sensitivity resource for each feature family, with its coverage across the ten languages the structural sample showed. The ten languages are Dutch, English, French, German, Greek, Italian, Korean, Polish, Portuguese and Spanish. Four rules govern that table and are fixed before any feature is computed. The first addresses a problem specific to multilingual work: a variable measured by a different instrument in each language confounds the method of measurement with the language, so that a language-level slope no longer means the same thing twice over. Each feature will therefore be computed from a single resource applied identically in every language, even where a better native resource exists in one of them, and native norms will be run as a pre-specified sensitivity analysis with the per-language correlation between the two reported. Second, no value is ever carried across languages by translating a word list, and a predicted resource is admitted only where the prediction was made within each language by one documented procedure and its correlation with native ratings is reported. Where no acceptable measure exists for a language, that language does not contribute that feature. Third, every measure reports the proportion of tokens it matched, and a language in which a measure matches fewer than 80 per cent of tokens does not contribute it. A per-text rule under the dimension reduction governs which texts enter the component solution. Fourth, the full inventory, the software versions, the model checkpoints, the tokenisation rules and the matched-token proportions will be frozen and

published before any feature is joined to an outcome. Korean is the language these rules constrain most, and the consequences for it are set out beside that table.

Measures

The processing outcome, serving Hypotheses 1 to 4 and 6, is the natural logarithm of elapsed reading time divided by the story's word count, standardised within reading language. We describe it throughout as reading pace or processing time, never as a measure of comprehension or fluency, because nothing in these data licenses that reading. The outcome for Hypothesis 5 is the child's optional post-reading rating of the story as easy, just right or difficult, treated as ordered and modelled on a cumulative-logit scale.

The confirmatory predictors are the three component scores defined below. Text length in words, narrativity, mean valence and mean arousal are pre-specified secondary text predictors and enter every model as covariates. The variability of valence and of arousal, and the additive and temporal connective densities, will be computed for description only: they enter no confirmatory model and are excluded from the component solution. Whether the text was generated or drawn from the classic set enters as a covariate, and the substantive comparison between the two is a contingent hypothesis, for reasons given under Table 2. Because predictors are standardised within language, a pooled coefficient is an average over languages of slopes each expressed in that language's own standard-deviation units. We will report the raw-scale standard deviation of each component per language so that readers can translate, and the language-varying slopes summarise the heterogeneity this induces.

Three variables describe the reader and the reading context. The age band is the platform's own record of the child's age group, reduced to the two bands this study covers. The band is taken per

session. An account whose recorded band changes within the window contributes each session under the band recorded for it, and the number of such accounts will be reported. The reading level is the level the family has set for the session, which determines which version of a text is served and is therefore an attribute of the reading as well as of the reader. Narration is the session-level flag recording that a spoken rendition accompanied the reading. All three are chosen by the family and none is randomised.

Accumulated in-app reading, which serves Hypothesis 6, is the cumulative count of words in completed distinct stories the account read before the current session, on a log scale. Session order, the number of days since the account's previous eligible session and the number of days since the account's first eligible session enter alongside it, so that accumulated reading is separated from the position of a session in a sequence, from recency and from the time elapsed since entry. Within tier 2 the measure is decomposed into the account's own mean and the current session's deviation from it, and the deviation term is the registered test.

The data are crossed, and not nested: readings sit within accounts, a text is met by many accounts and exists in several languages and at several reading levels, and an account may read in more than one language. We will therefore treat Account, Text, Language, Country and App version as crossed grouping factors, and report the counts supporting each before fitting any model. A grouping factor other than Text and Account enters when it has at least five levels with at least 20 eligible readings each. Below that it is dropped, and the drop will be reported. Reading level, with only three levels, enters as a fixed factor. If at least 90 per cent of texts are read in only one eligible session, a Text varying intercept is not estimable and is replaced by varying intercepts for the repeated attribute combination that generated the text. These thresholds are registered conventions,

and any consequence of them will be reported.

Randomisation and blinding

Nothing in this study is randomised. It is a secondary analysis of records generated by families using a commercial application, and no allocation, assignment or condition is under anyone's control. The only random assignment anywhere in the protocol is the permutation quality check, where we reassign texts to readings at random precisely to destroy the association under test.

Data collection and analysis will not be performed blind to the conditions of the study. Data collection is performed by the platform without researcher involvement. Before in-principle acceptance, blinding is structural: we cannot see the outcome data because the data controller holds them, under the sequestration arrangement described above. After transfer, analysis will not be blind to the values of the predictors or grouping variables, since the registered models require them. We will write, run and debug the outcome-model code against data simulated from the design specification deposited with this protocol, so that every script is complete and exercised before the real file exists on any machine of ours.

Sampling plan

The design is observational and the sample is a census: we will analyse every eligible record in one immutable extract, whose closing date is fixed before any outcome model is fitted, with no sequential testing and no optional stopping.

Two nested samples serve different hypotheses, because those hypotheses are identified at different levels and the data supporting them are unequally certain.

Tier 1 carries the text hypotheses. It comprises every completed reading in the extract for which

the account is recorded in age band 7-8 or 9 and over, the reading language is one the text-feature pipeline covers, the story text has been supplied, and the reading time falls inside the plausibility bounds below. Where an account read the same text more than once, only the first reading enters tier 1, and repeated readings are retained for the practice analysis described under sensitivity. The unit of analysis is one reading of one text by one account.

Whether tier 1 can estimate a child varying intercept depends on how many accounts read more than one distinct text, which is not known in advance. If at least ten per cent of tier-1 readings come from accounts contributing two or more readings, the model includes a varying intercept for Account. If fewer do, that term is not separable from residual variance: it is dropped, child variance is carried in the residual, and the reduction in the effective independence of the sample is stated wherever a tier-1 interval is reported. The ten per cent figure is a registered convention. We will read the proportion from the provider-computed counts before fitting anything.

Tier 2 carries the within-child hypothesis. It comprises those tier-1 accounts contributing at least three eligible readings of distinct texts. Three is the smallest number that leaves a degree of freedom once a within-child slope and an intercept are estimated, and the threshold will be varied to two and to five in the sensitivity analyses. Tier 2 estimates a child varying intercept and separates each child's mean level of a predictor from that reading's deviation from it, so a within-child association is not conflated with stable differences between children who habitually read at different rates. How many accounts reach the tier-2 threshold is not known in advance. The structural sample contains none, but it holds one session per account and cannot speak to the full window.

Reading time enters as elapsed seconds for the whole story divided by the story's word count, and we bound it by what a child could physically be doing, and not by anything observed in these data.

We will exclude a reading faster than 400 words per minute, or 0.15 seconds per word, as faster than reading. We will exclude a reading slower than 10 words per minute, or six seconds per word, as a story left open. The upper bound follows a published precedent that states such a rule and its yield: a meta-analysis of Spanish reading rates deleted silent rates above 400 words per minute, on the ground that such a rate exceeds what a competent adult can manage and indicates skipped text (Rayner et al., 2016), removing 313 of 19,237 cases (Ripoll Salceda et al., 2020). The lower bound sits below the slowest pooled mean in that meta-analysis, 30 words per minute for Spanish first-graders reading silently, though not below the floor of the normal range proposed for that grade, 8 words per minute. From second grade the range begins at 41. A floor of 10 words per minute can therefore exclude a genuinely slow reader only among the youngest first-graders, most of whom the age criterion removes in any case. Both bounds are wide by design, so the primary analysis removes little beyond the impossible. Tighter treatments are sensitivity analyses. We will report the number and proportion removed by every rule, by language, age band, reading level and narration state.

One further rule concerns the top of the distribution. Where a session timeout caps the recorded interval, the cap shows itself as an implausible number of sessions sharing one value, which is one of the quantities the provider-computed counts report. If those counts show more than one per cent of sessions sharing a single value, we will locate that value in the transferred extract before fitting any outcome model. If it lies in the upper decile of the reading-time distribution, we will treat it as a right-censoring point and fit the timing models as censored lognormal models, with those observations censored, and not discarded.

Ahead of all of the above, we will exclude sessions the platform flags as incomplete or crashed, duplicated event records, accounts flagged as duplicates under the platform’s own data-quality

rules, and stories without a positive word count. No excluded datum is replaced. A participant-flow diagram will report every exclusion and the retained accounts, readings and texts, overall and by language.

Measurement, availability and missingness audit

Measurement error in digital traces has its own taxonomy, and the errors specific to platform data are not the errors of surveys (Sen et al., 2021). Of the error types that framework names, four apply here. The platform defines the categories, so age band and reading level are its constructs. The instrument drifts, because the app changes version across the window. The interval that the timing column covers is not fully specified, which is question (c). And a session timeout would right-censor the outcome, which is why the censoring rule is registered in advance. None of these is fatal, and each is one of the reasons the audit precedes the models.

Before fitting any outcome model, we will run and archive an audit of the frozen extract and its dictionary. It establishes field availability and units, the granularity of the age field, how often a text and a reading level recur, the narration flag's behaviour within accounts and within sessions, duplicate and crash flags, changes across app versions, per-language feature coverage, and every nesting and crossing count the models require. It also reports missingness for every planned variable, overall and by language, age band, reading level, narration state and session order.

Three rules govern what the audit may change. A feature that is structurally unavailable in a language is not imputed: the language simply does not contribute it. A session missing an outcome is excluded from the model for that outcome alone, and a missing narration flag does not remove a session from a model that has no narration term. No variable is selected, transformed or discarded because of its association with reading time or with the rating.

The rating in Hypothesis 5 is optional and was absent for 25 of the 35 sessions in the structural sample, so high missingness is expected. Where the rating is missing for more than five per cent of otherwise eligible sessions, we will model availability from observed pre-response variables and report an inverse-probability-weighted analysis as a sensitivity check, with weight distributions and common support shown.

Design and precision analysis

Nature Human Behaviour's guidelines set 0.95 as the a priori power floor for frequentist plans and, for Bayesian plans, allow a maximum feasible sample provided an inconclusive result at that size would still be an important message. This design is neither a significance test nor a Bayes-factor test but an estimation design on a census of existing records, and it maps onto the second provision. The frozen extract is the maximum feasible sample, since no further data can be collected, and 0.90 is registered as the assurance level at which a hypothesis counts as adequately powered, with every 0.95 requirement reported beside it from the same archived curves (Supplementary Section S5) so that a reader can apply the stricter target. An inconclusive verdict at the census is itself informative. If the extract holds too few distinct texts, we will say so before fitting and report that this platform cannot answer the question, which matters for a class of data source that others will be tempted to use. A hypothesis declared underpowered is still fitted and reported, so its interval bounds the effect even where it cannot decide it.

Because we will analyse every eligible record, the question is not how large a sample to collect but what the available records can settle. Here that turns on a single quantity: the text coefficients are identified between distinct texts, and how many distinct texts the extract holds is unresolved.

We therefore simulated data from the design specifications deposited alongside this protocol with

pilotr, sweeping the number of distinct texts at constant readings per text and separately sweeping readings per text at a constant number of texts (Bernabeu, 2026b). The simulated tier-1 model contained the three components, the age band, the two age-band interactions and a text intercept, fitted by maximum likelihood. The narration model contained the lexical and continuity components, the narration flag and their interactions with it. The tier-2 model added a child intercept and a single accumulated-reading trend. The simulated models therefore omit the narration terms of the tier-1 model, the reading-level adjustment, the covariates, the language, country and app-version intercepts, the language-varying slopes, the distributional sub-model and the review-derived priors that the registered models carry. Simulation was on the within-language standardised scale, with the text intercept contributing 9 per cent of the variance, the child intercept 20 per cent and the residual the remainder, so a simulated coefficient is directly comparable with the region of practical equivalence. The simulation generated child variance and then handed the analysis a dataset in which each account contributes one reading, so that variance reached the fitted model as inflated residual, and in that one respect the requirements below are an upper bound on what the design needs. The omissions do not all point the same way. Adjusting for reading level, which moves all three feature families together, will remove some of the components' variance and raise every requirement, whereas the language-varying slopes and the priors will lower them. The simulation balanced the age band and narration, and the main effects it priced are balanced averages. The registered prevalence-weighted main effects carry the precision of the pooled slope and no design effect, whereas each interaction inherits the design effect for imbalance that the adjustments below apply. We will refit one grid point of each sweep with the registered brms specification when the design analysis is re-run at the observed inputs, and report the difference. A coefficient the size of the region can never be declared determinately outside it, so we ran the sweep at twice that size, the

smallest effect this protocol treats as worth reporting. A null condition with every coefficient set to zero established how reliably the design can declare practical equivalence, so an inconclusive verdict is predictable in advance.

The region of 0.05 standard deviations and the 0.10-standard-deviation effect are registered conventions, the second being twice the first because an effect at the boundary can never be declared outside it. The accompanying review supplies the check on them. The prior-derivation pipeline (Supplementary Section S6) will record, before any outcome datum is accessed, whether each component's pooled estimate on the model's scale exceeds 0.10 standard deviations. Where a pooled estimate lies between 0.05 and 0.10, the texts sweep will be re-simulated with the deposited scripts at that effect, replacing the single 0.10 input, and the resulting threshold replaces the one in Table 1 for that component before fitting. Where a pooled estimate lies at or below the region's half-width, no number of texts could yield a determinate verdict at that size. The 0.10 convention then stands, and the discrepancy will be reported beside the component's verdict.

For the three text coefficients, an effect of 0.10 standard deviations reached a determinate verdict nine times in ten at 738 distinct texts for lexical difficulty (95 per cent interval 640 to 843), 660 for grammatical load (593 to 731) and 634 for referential continuity (568 to 704), and 95 times in 100 at 926, 812 and 784. The interactions with the age band needed roughly twice as many: 1,230 texts for grammatical load (1,107 to 1,359) and 1,431 for referential continuity (1,276 to 1,596) at 0.90, and 1,510 for grammatical load at 0.95, a target the continuity interaction did not reach within the 1,600 texts simulated. The narration contrast needed 1,234 texts (1,113 to 1,361) before the design effect for imbalance is applied, and 1,506 at 0.95. That figure prices the lexical-by-narration interaction coefficient, and is a lower bound for the difference-of-proportions rule Hypothesis 4

registers, whose draw-by-draw ratio has heavier tails than the coefficient it is built from. Declaring practical equivalence when there is nothing there cost about the same: at 800 texts a main effect was correctly declared equivalent in 87 to 90 per cent of replicates and at 1,600 in every one, while the interactions reached 89 to 92 per cent only at 1,600, over 120 replicates. An inconclusive verdict is therefore predictable from the number of texts alone. The within-child term needed 563 children in tier 2 (463 to 674) contributing five readings each, and 681 (543 to 834) at 0.95, for the unadjusted trend. The registered term is adjusted for session order and elapsed time, with which accumulated words are correlated within a child, so this figure understates the requirement, and the tier-2 re-run described below re-derives it with those covariates in place.

The second sweep is the more instructive. Holding the number of distinct texts at 250 and raising readings per text from 2 to 60 narrowed the interval on a main effect from 0.19 to 0.08 standard deviations, yet never carried it to a determinate verdict, reaching 0.70 at best. Over the same range the age-band interactions climbed from under 0.11 to between 0.87 and 0.90. Sessions and texts are therefore not interchangeable: a text coefficient is bounded by how many different texts the platform generated, whereas an interaction between a text property and a reader attribute gains from many different children meeting the same text. This matters because sessions are the quantity this platform has in abundance and distinct texts are the quantity in doubt, so a very large export could still leave Hypothesis 1 underpowered while answering Hypotheses 2 and 3.

Figure 3 shows both sweeps. The threshold each hypothesis has to clear is recorded in Table 1, and we fixed it before the extract was released. A hypothesis whose required contrast is absent will be labelled untestable, and one whose simulated interval cannot separate the region’s boundary from the effect will be declared underpowered, before its model is fitted.

Five adjustments of the design analysis are registered, each a function of the provider-computed counts or of the archived prior pipeline. Two are analytic. The simulation balances the narration contrast, whereas narration was on in 32 of the 35 sessions of the structural sample. An unbalanced binary contrast at prevalence p inflates the variance of its interaction coefficient by $1/(4p(1-p))$ relative to balance, so we will multiply the simulated Hypothesis 4 requirement by that factor at the observed narration prevalence and state the adjusted requirement before fitting. At a narration prevalence of 0.91, equivalently an unnarrated share of 0.09, the factor is about 3.0. The second adjustment applies the same factor to the Hypothesis 2 and 3 requirements at the observed age-band prevalence, since the simulation balanced the bands as well. Two are re-runs of the deposited scripts. The rating simulation will be re-run at the observed rating availability, replacing that single input. The tier-2 sweep will be re-run at the observed distribution of sessions per child, and for that re-run alone the deposited tier-2 specification will be extended with session order and days since the first session, with per-session word counts generated so that accumulated words correlate with session order as the counts show. The extended specification will be deposited before it is run. The thresholds those re-runs yield replace the ones in Table 1 for the underpowered declarations of Hypotheses 5 and 6. The fifth adjustment is the re-simulation at a pooled effect below 0.10 described above, which runs only if the prior pipeline calls for it.

Analysis plan

We will fit all confirmatory models as Bayesian hierarchical models with the brms package and its Stan backend (Bürkner, 2017), standardising continuous predictors within language after transformation. Wherever the accompanying systematic review supports it, weakly informative priors will be replaced by priors derived from that review, as described under Priors.

Outcome-blind dimension reduction

As recorded under Data provenance, none of what follows has been run, and the first fitting will follow in-principle acceptance. The observed text features are correlated because the same decision about what reading level a text is written for moves lexical, grammatical and discourse properties together. We will use principal component analysis to reduce them without treating them as effects of an unobserved latent cause (Crossley, 2025), and will conduct it separately within the lexical, grammatical and discourse families. The affective measures will be kept as separate secondary predictors, because valence and arousal are theoretically distinct and need not move together. Narrativity, though computed as a discourse feature, will likewise be excluded from the component solution, because it enters every model as a separate secondary predictor and must not sit on both sides. Each distinct text will contribute one row regardless of how often it was read, so a popular text does not determine the solution, and a text enters a family's solution only if it has valid values on every retained feature of that family.

Counts will be transformed with $\log(1 + x)$ and bounded proportions with an empirical-logit transform where these reduce skew. A feature is dropped from a language if more than 20 per cent of values are missing, if fewer than 80 per cent of texts receive a valid token-level match, or if its standard deviation after transformation is below 0.01. A language contributes to a family's solution only if it has at least 100 distinct eligible texts and at least three valid features in that family, two of them anchors. Where two features correlate at $|r| > .90$, the retained one is chosen in this order: lower missingness, then broader language coverage, then higher extraction reliability on a manually checked sample, then the measure listed first under Materials. No outcome correlation enters any of these decisions.

Features will be standardised within language. To stop the largest language determining the solution, we will compute a correlation matrix within each eligible language, transform the correlations to Fisher's z, average the matrices with equal weight per language, and transform the average back before the analysis (Revelle, 2023). Sampling adequacy requires individual Kaiser-Meyer-Olkin values above .50 and an overall value above .60. The number of components will be set by parallel analysis against the 95th percentile of 10,000 random datasets (Horn, 1965). The scree plot will be reported but does not override that rule. Where a family retains more than one component, we will use promax rotation with $\kappa = 4$, because linguistic properties may correlate (Biber, 1988). We will then apply the regression scoring coefficients from the frozen, language-balanced solution to each text's within-language standardised features.

Each confirmatory component must load at least .30 in absolute value on at least two of its anchors. For LexicalDifficulty, the anchors are frequency, surprisal and word length. For GrammaticalLoad, they are dependency distance, subordination, clause density, sentence length and noun-phrase depth. For ReferentialContinuity, they are adjacent-sentence similarity, referential overlap and the density of causal and contrastive connectives. Age of acquisition contributes to the lexical family where a broad native norm set exists but is not an anchor, because the cross-linguistic set holds only 299 words per language. Only the causal and contrastive connective densities enter the discourse family, and both are anchors. The additive and temporal densities are descriptive only, as stated under Measures. The restriction is substantive: in 794 readers across 20 texts, causal and contrastive connectives raised comprehension, temporal ones did nothing and additive ones lowered it, so a single mixed count has no defined expected sign (Kleijn et al., 2019). Signs are fixed so that higher LexicalDifficulty and GrammaticalLoad mean a more demanding text and higher

ReferentialContinuity a more explicitly connected one. If no component in a family satisfies the rule, the hypotheses resting on that family are reported as untestable. The complete pattern matrix, the component correlations, the variance explained, the language-specific matrices and the loading congruence for the five largest languages will be reported. Language-specific and sparse solutions are sensitivity analyses, and never alternative routes chosen after seeing an outcome.

Confirmatory models

In tier 1, we will estimate a model of within-language standardised log reading time per word with the three components, the age band, the narration flag, the interactions of each component with the age band and with narration, the reading level, log text length, narrativity, valence, arousal and the generated-or-classic indicator as fixed effects, with the age band and narration entered as deviations from their observed prevalence as stated in the Introduction. Its grouping structure comprises varying intercepts for Text, Country and App version, varying intercepts and uncorrelated component slopes for Language, and a varying intercept for Account where the estimability rule above is met. Hypotheses 1 to 4 will be evaluated in this model. The slope of a component in a given state is recovered draw by draw as the main effect plus the interaction multiplied by that state's deviation score, so that the unnarrated slope is the main effect plus the narration interaction times the unnarrated deviation score, which is zero minus the narration prevalence, and the narrated slope is the main effect plus the interaction times one minus the prevalence.

The Hypothesis 4 contrast will be computed as follows. For each posterior draw, the proportional attenuation of a component is one minus its narrated slope divided by its unnarrated slope, so that a positive value means a smaller coefficient under narration. The gate registered in the Introduction is evaluated on the derived unnarrated slopes: the contrast is computed only when fewer than 5 per

cent of the draws of each unnarrated slope lie inside the region of practical equivalence. Draws in which either unnarrated slope lies inside the region will be excluded from the ratio, and the excluded fraction will be reported. As a registered sensitivity analysis, the contrast will be repeated with a fixed denominator, each narration interaction divided by the posterior median of its unnarrated slope, and reported beside the draw-by-draw version.

Three further rules govern Hypothesis 4. First, the within-account contrast, restricted to accounts observed in both narration states, becomes the primary Hypothesis 4 estimate if those accounts cover at least as many distinct texts as the adjusted requirement, with the full-sample contrast reported beside it. Otherwise the within-account contrast remains a sensitivity analysis, and the selection into narration state is named wherever the full-sample estimate is reported. Second, the tier-1 model carries a distributional sub-model in which the residual standard deviation is itself predicted by the narration flag, and the ratio of the narrated to the unnarrated residual standard deviation is the compression diagnostic. A ratio below 0.5 is the registered signature of a recording setting the pace. If it fires, the full-sample contrast is not interpreted and Hypothesis 4 rests on the subset described next. Third, the contrast will be repeated in the narrated sessions the audio cannot have paced. For each language, the narration rate will be estimated as the median words per minute of its narrated sessions, and a narrated session counts as unpaced when its rate exceeds 1.5 times that median. Where a language has fewer than 100 narrated sessions, a fixed bound applies instead: that language's own adult read-aloud mean in the International Reading Speed Texts where the language is among the 17 covered, and otherwise the slowest of those means, 138 words per minute for Arabic (Trauzettel-Klosinski & Dietz, 2012). The premise, stated here so that the audio duration would let it be checked, is that a narration recorded for children is paced no faster than adult oral reading in the

same language. If the selectivity survives where the recording was not the constraint, pacing does not explain it. Hypothesis 4 is reported as underpowered from the subset if the subset holds fewer distinct texts than the adjusted requirement. Should the data controller supply the audio duration, these rules are replaced by the duration itself and we will report the substitution as a deviation from this protocol.

The Hypothesis 5 model will have the same fixed and grouping structure and a cumulative-logit likelihood over the three rating categories, with category thresholds allowed to vary by language so that a difference in how families in different countries use the middle category is not read as a difference in text difficulty. A text rated by fewer than five distinct children does not contribute, and the primary model is complete-case, with the inverse-probability sensitivity analysis registered under the audit.

For tier 2, we will refit the same specification with a varying intercept for Account, decomposing accumulated reading into the account's mean and the current session's deviation from it, alongside session order, days since the previous session and days since the account's first eligible session. The deviation term is the registered test for Hypothesis 6.

Posterior predictive checks will assess calibration and residual structure by language, age band, reading level and narration state in every model.

Priors

For each confirmatory main effect, and for the secondary text predictors where the review holds at least two studies for them, priors will be derived from the accompanying systematic review through a deterministic pipeline whose rules we froze before any extraction row existed, and which will be run, with its output archived, before any outcome datum is accessed. The review's extracted

estimates for the corresponding predictor family and outcome class will be sign-harmonised so that facilitation always means faster reading, converted to a common standardised scale, and pooled by random-effects meta-analysis with one estimate per study per relation. The prior is Student's $t(3, \hat{\mu}, \sigma)$, where $\hat{\mu}$ is the pooled estimate translated to the model's scale and σ is twice the combined pooling and between-study uncertainty, with a floor of 0.10. The doubling and the floor guard against the gap between a laboratory comprehension test and a passively timed reading in a game.

The same review motivates the hypotheses, so the evidence that generates a prediction also centres its prior, and a reviewer is entitled to ask whether that makes a predicted verdict easier to reach. Inference here is estimation against a region, and not a Bayes factor, so a prior centred in the predicted direction does not enter the decision as a hypothesis weight. The doubled scale and heavy tails make the derived priors weakly informative relative to the likelihood that hundreds of texts will supply. And we will report prior-data conflict diagnostics for every review-derived prior. We register the tie-break as well: the fit with the derived priors carries the registered verdict, and a hypothesis whose label changes under the weakly informative refit described next is reported as sensitive to the prior and is not counted as a determinate confirmatory verdict.

Coefficients without adequate review evidence, which means every interaction term and any family supported by fewer than two studies, keep weakly informative priors: $t(3, 0, 0.5)$ for standardised main effects and $t(3, 0, 0.25)$ for interactions. Group-level and residual standard deviations take half-$t(3, 0, 1)$ priors and any estimated correlation matrix takes LKJ($\eta = 2$). Intercept priors will be calibrated by prior-predictive simulation to the outcome scale, without reference to any predictor-outcome association. The derivation rules are summarised in Supplementary Section S6, and the full prior table with its forest plots will be archived before the extract is transferred and will accompany

the Stage 2 report. We will refit every model with weakly informative priors throughout, and with the derived scales halved and doubled, and report conclusions under each.

Decision rules and diagnostics

For every confirmatory coefficient, we will report the posterior median, the 95 per cent highest-density interval, the probability of the predicted direction, the proportion of the posterior inside the region of practical equivalence, and a model-implied contrast on an interpretable scale. The region is $|\beta| < 0.05$ standard deviations for a one-standard-deviation change in a predictor of reading time, and a half-width of log(1.05) on the log-odds scale for the rating model. More than 95 per cent of the posterior inside the region is read as practical equivalence, and less than 5 per cent inside it as a practically non-negligible association, labelled as in the predicted direction or in the opposite direction according to the sign of the posterior median. Everything else is inconclusive. For the ordering clause of Hypothesis 5, a posterior probability of at least 0.90 that the three coefficients rank as they do for reading time counts as agreement on order, and anything lower as order not established. For Hypothesis 4, at least 95 per cent of the posterior of the difference of attenuations above 0.20 supports selective removal of decoding cost, more than 95 per cent inside the region supports audio pacing, more than 95 per cent below -0.05 is reverse selectivity, and anything else is inconclusive. The fixed-denominator sensitivity contrast is read against the same thresholds.

Each hypothesis is a single pre-registered test with its own decision rule, and no claim in this protocol aggregates evidence across hypotheses, so no correction for multiple comparisons will be applied. The one composite claim, the overall Hypothesis 1 verdict, is an intersection-union rule: it is recorded only when all three coefficients are individually non-negligible in the predicted direction, so its error rate cannot exceed that of any single component. Hierarchical shrinkage and

the null condition of the design analysis, which reports how often a zero effect is wrongly declared non-negligible, calibrate the per-decision error rates behind Table 1.

Four registered contingencies read fitted quantities after fitting, and none of them changes whether a hypothesis is tested, on which sample, or with which model. The first is the Hypothesis 4 compression diagnostic, which withholds the full-sample contrast. The second is the exclusion of draws with an unnarrated slope inside the region from the Hypothesis 4 ratio. The third is the third outcome-neutral check, which makes the within-level estimate primary for a component whose pooled coefficient it halves. The fourth is the prior-sensitivity label above. Each is stated where it applies, and each is an IF-THEN rule fixed here.

We will run four chains of 4,000 iterations, half of them warm-up. Acceptable sampling requires $\hat{R}$ below 1.01, bulk and tail effective sample sizes above 1,000 for every confirmatory parameter, no divergent transitions, satisfactory energy diagnostics and no unresolved maximum-treedepth warnings. Models failing these will be refitted with a higher target acceptance rate, greater tree depth and more iterations. If simplification is still needed, language-level slopes will be removed in the fixed order lexical, then discourse, then grammatical, and every estimable varying intercept will be retained. Every simplification and its diagnostic consequence will be reported. Variance inflation factors will be reported for the fixed-effect design matrix. A high value does not trigger an outcome-dependent deletion: the joint model remains primary, and single-family models will be reported as a collinearity sensitivity analysis.

For Hypotheses 2, 3 and 4, we will compare the model with one omitting the relevant interaction, using Pareto-smoothed importance-sampling leave-one-out cross-validation, and report the expected log pointwise predictive density difference, its standard error and the Pareto-k diagnostics, with

exact refits where importance sampling is unreliable. Predictive comparison supplements the coefficient-level summaries and does not gate them.

Outcome-neutral quality checks

Five checks will run before the confirmatory models are interpreted, each with a stated pass criterion and a stated consequence of failure, and a sixth, the ordering clause of Hypothesis 5, is evaluated with that hypothesis. The first, second, fourth and sixth are decided without reference to any confirmatory coefficient. The third is a robustness condition computed from the fitted text coefficients, and the fifth is a registered clause of Hypothesis 5.

The first is a permutation null. We will reassign texts at random to readings within language and reading level, and refit the tier-1 model 200 times. The refits will use the maximum-likelihood analogue of the tier-1 model, with the same fixed and estimable grouping structure, and the region will be applied to the 95 per cent Wald interval, since 200 full Bayesian refits are not feasible and the check concerns leakage rather than calibration. No text coefficient may be declared practically non-negligible in more than 5 per cent of those refits. An inconclusive verdict on permuted data is the expected outcome at moderate text counts and does not count as failure. A coefficient that survives permutation means the pipeline is carrying structure that does not come from the pairing of a text with a reading, and we will withhold the confirmatory results until the leak is found and reported.

The second is a positive control on the timing column. Total elapsed time must rise with the story's word count, with more than 99 per cent of the posterior above zero. This is close to a tautology if the column means what the dictionary says it means, which is why its failure would be informative: a timing field dominated by idle or backgrounded time need not track how much text there was to

read. Failure means the timing column does not index reading, and Hypotheses 1 to 4 and 6 are reported as not evaluable on this extract.

The third asks whether the association is an artefact of platform assignment. We will re-estimate the three text coefficients within each reading level separately, and each must retain at least half the magnitude it had in the pooled model. The account on which text difficulty is merely assigned to children who differ in capacity predicts that these coefficients shrink towards zero when the level is held fixed. The account this protocol advances predicts that they do not. This check is a robustness condition and not a decision about testing: if a coefficient falls below half its pooled magnitude, the within-level estimate becomes the primary estimate for that component and the pooled one is reported beside it, with the assignment confound named wherever the coefficient appears.

The fourth is a set of three independence claims the theory object entails. Conditional on the reading level and on whether a text was generated or classic, the three components should be close to independent across texts. Narration should be unrelated to accumulated reading. And the difficulty rating should carry no further information about reading level once reading time is conditioned on. We will report each as an estimate with an interval, and an implication is judged to have failed when the 95 per cent interval of the corresponding conditional association lies entirely beyond 0.10 in standardised units. A failure does not invalidate the confirmatory tests, but it identifies a path the causal diagram omits, and it is reported as such.

The fifth is convergent validity, the ordering clause of Hypothesis 5, specified there and evaluated with it, with its outcome under an underpowered Hypothesis 5 stated there.

The sixth is a validation of the feature pipeline itself. We will check a stratified random sample of 50 texts per language by hand for tokenisation, sentence splitting and dependency attachment, and

report the agreement statistics per language before any feature is joined to an outcome. A language whose labelled-attachment agreement with the hand-checked parse falls below .85 contributes to descriptive reporting but not to the grammatical component.

Sensitivity analyses

The following sensitivity analyses are pre-specified. We will refit the timing models with no plausibility bounds at all, with winsorisation at the 1st and 99th and at the 0.5th and 99.5th percentiles within language, and with posterior-predictive residual trimming. Where the censoring rule is triggered, we will report the censored and uncensored fits side by side. We will vary the tier-2 threshold from three distinct texts to two and to five, and add repeated readings with a repetition indicator. We will restrict analyses to the five largest languages, and fit language-specific models and component solutions with loading congruence reported. We will replace the component scores with sparse-component scores and with pre-specified equal-weight composites of the anchor features. Where the audit finds the age field finer than two bands, we will re-estimate the Hypothesis 2 and 3 interactions with age entered as an ordinal monotonic predictor and report the result beside the two-band contrast. For narration, we will report the within-account contrast where it is not already primary, and the fixed-denominator contrast registered under Confirmatory models. We will compare the rating model under adjacent-category and cumulative-logit links, stratify it by language, and compare the inverse-probability analysis for rating non-response with the complete-case estimates. We will compare Text grouping with grouping on the attribute combination that generated the text. Finally, we will halve and double the prior scales and apply the alternative regions of practical equivalence, 0.03 and 0.10 standard deviations for reading time and half-widths of log(1.03) and log(1.10) on the log-odds scale for the rating.

To ease comparison with the prior literature, we will report maximum-likelihood mixed models with the same fixed and estimable grouping structure in a supplementary table. They are descriptive, carry no confirmatory weight, and will not be used to resolve a Bayesian result.

What is not yet settled, and what follows from each answer

We put thirteen questions about the extract to the data controller on 7 August 2026, and they remain open at submission. Leaving them open is not a gap in the protocol so long as each has a decision rule attached in advance, and Table 2 attaches one to each. Every rule below is a function of the answer alone.

Five of these questions correspond to auxiliary assumptions recorded in the theory object, which is how the protocol keeps the statistical contingency and the theoretical commitment in the same place. If the answer to a question falsifies its assumption, the predictions that assumption protects are the ones the table reassigns.

Table 2

Questions outstanding at submission and the decision rule attached to each

What is unresolved	What it bears on	What the protocol does under each answer
How many distinct texts carry eligible readings, settled by the provider-computed counts together with question d. The controller's own estimate spans a few hundred to a few million, according to whether one combination of chosen attributes yields one shared text or a fresh text per session; the published pre-launch description is a catalogue of over 40,000 generated stories.	Every text coefficient, since all are identified between texts.	The count will be read from the provider-computed figures before modelling and located on the precision curve in Figure 3. Below the threshold in Table 1 a hypothesis is declared underpowered before its model is fitted. Where each session carries its own text, the Text varying intercept is replaced by varying intercepts for the repeated attribute combination, under the sparsity rule given under Measures, and precision is then governed by the number of attribute combinations. We will report which case obtains.
Whether the narration flag records that audio played or only that it was available, and whether it can change within a session (question a).	Hypothesis 4, through the auxiliary assumption that narration played.	If it records playback, Hypothesis 4 stands as written. If it records availability only, the variable is relabelled as availability, the contrast becomes an intention-to-treat comparison whose attenuation is diluted towards the null by non-compliance, and that dilution is stated wherever the coefficient is reported. Sessions in which the flag changes are excluded and counted.
Whether one story identifier maps to one text, and whether a text differs across reading languages by translation or by independent generation (question d). The published pre-launch description says many stories were translated from English.	The identity of the stimulus, through the auxiliary assumption that the keyed text is the text the child read, and therefore every model's grouping structure.	If one identifier is one text, texts are counted by identifier crossed with language and reading level, as specified. If a text is translated across languages, a varying intercept for the underlying story is added, crossed with Language. If each language is generated independently, texts in different languages are distinct and no such term is added.
Whether a stable identifier exists for each session (question e). The sample carries none.	The join between the extract and the texts, and the check of the transferred file against its manifest.	If one exists, it is carried in the export and used for the join. If none exists, sessions are joined to texts on story identifier, reading language, reading level and app version, as Data provenance states, and the transferred file is checked against its manifest by row count and checksum. Neither answer changes a model.

What is unresolved	What it bears on	What the protocol does under each answer
What the reading-time interval covers: whether it includes embedded activities, and whether it includes time backgrounded or idle (question c).	The outcome itself, through the auxiliary assumption that elapsed time indexes effort.	If the interval excludes them, the outcome stands as written. If it includes embedded activities, texts that offer them are identified from the story attributes, an indicator enters every model, and the primary analysis is repeated within texts that offer none. If it includes idle or backgrounded time, the upper plausibility bound and the censoring rule carry more of the work, and the limitation is stated in the abstract itself.
Whether an account persists across reinstalls and whether siblings can share one (question f).	Tier 2 and Hypothesis 6, through the auxiliary assumption that an account is a child.	If an account can be shared, the within-child reading of Hypothesis 6 is withdrawn in advance and the coefficient is reported as a within-account association, with the ambiguity stated in the hypothesis itself.
Whether the reading level can change within an account over time, and whether it changes the text served or only the page count (question g).	The platform-assignment confound and the third quality check.	If it changes only pagination, texts are comparable across levels and the pooled estimate is primary with the within-level estimate as a check. If it changes the text, the within-level estimate becomes primary and the pooled one is reported alongside it. A within-account contrast on a level change is not registered here; any such analysis would appear at Stage 2 under Exploratory analyses.
The source of the age band and how much of it is missing across the whole window (question h). It was blank for 10 of 35 sessions in the structural sample.	Hypotheses 2 and 3.	Missingness will be reported by language, reading level and app version. Above 20 per cent, availability is modelled from observed pre-response variables and an inverse-probability-weighted fit is reported alongside the complete-case fit, with both interpreted together.
Which reading languages the window covers. The app is marketed in 19 languages and the structural sample showed ten (question i).	Coverage of the feature pipeline and the language grouping factor.	The observed set will be reported as found. A language for which the required resources do not exist contributes to the participant-flow diagram and the descriptive tables but not to any feature-dependent model, and the excluded volume is quantified.
Whether the classic stories are fixed, human-authored texts identical for all users within a language (question j).	The comparison between machine-generated and human-authored material, contingent hypothesis C4.	The condition was not confirmed before submission, so C4 is not activated. The generated-or-classic indicator remains a covariate, and if the condition is confirmed later the comparison would appear at Stage 2 under Exploratory analyses.
Whether unfinished sessions are retained anywhere, and what the completion rate is (question k).	Generalisability, since every row in the extract is a completed reading.	The study describes completed readings and says so. Where an aggregate count of unfinished sessions by language is available, the completion rate is reported and the restriction quantified.

What is unresolved	What it bears on	What the protocol does under each answer
What fields the older data carry and what window they cover (question l).	Hypothesis 6, through the within-window assumption that accumulated reading acts on decoding automaticity over the export window and not the reverse. The measure is truncated by a window opening on 1 January 2026.	If an older file carries the same fields, accumulated reading is computed over the longer window and the analysis is repeated on both. Otherwise the measure is left-truncated at the window's opening and is described throughout as reading accumulated within the observation window, which makes an experienced reader look new.
Whether post-story responses include comprehension items or only the difficulty rating (question m).	Contingent hypotheses C1 to C3.	The condition was not confirmed before submission, so C1 to C3 are not activated. If item-level responses turn out to exist, the analyses written for them would appear at Stage 2 under Exploratory analyses, and no comprehension claim is made among the confirmatory results.
What the per-session activity counters measure. In 6 of the 14 structural-sample sessions with activity, they sum to more than the whole reading duration, in one case 438 seconds against 222 (question b).	Nothing, by decision.	These fields are set aside and are not used as a predictor, a covariate or an exclusion criterion. The decision is recorded here so that their later use would be visible as a deviation.

Contingent hypotheses

We specify four further hypotheses in full here, outside the confirmatory set, because each depends on a fact about the extract that the data controller has not yet confirmed. Each carries an activation rule, and the rule is a function of that fact alone, read from the controller's written answer or, for the requirement that accounts be observed in both narration states, from the provider-computed counts.

A contingent hypothesis activates only if its condition is confirmed in writing by the data controller before this protocol is submitted, and the confirmation is reproduced in Supplementary Section S7. A condition confirmed after submission does not activate anything: the hypothesis is then reported at Stage 2 as exploratory, under that label, and is not counted among the confirmatory tests. This asymmetry is deliberate, because a hypothesis activated after review has not been reviewed. At submission, no condition has been confirmed, so none of the four is activated, and Supplementary Section S7 records that. They are retained here so that, should a condition be confirmed later, the analysis reported at Stage 2 under Exploratory analyses is the one written before any data were seen. A hypothesis whose condition is never confirmed is not run.

C1: coherence and inferential comprehension. Condition: item-level comprehension responses exist and carry a recoverable question type. We would predict that, in an item-level model of response correctness, ReferentialContinuity × QuestionType is positive with inference questions coded 1 and literal questions 0, so that the association of continuity with correctness is larger where the question requires integration (Aparicio et al., 2022; Freed & Cain, 2021; Pickren et al., 2022).

C2: narration and comprehension. Condition: item-level comprehension responses exist, and accounts are observed both with and without narration. We would predict that, within an account,

narration is positively associated with response correctness after text, age band, reading level, accumulated reading and language are adjusted for. The expected size is small (Clinton-Lisell, 2023; Wood et al., 2018), and the coefficient identifies an association, and never an effect.

C3: reading pace and comprehension. Condition: as C2, restricted to sessions without narration. We would predict that an account's mean log reading time per word is negatively associated with response correctness, so that accounts with a faster habitual pace answer more accurately (Sortkær et al., 2021). The simultaneous within-account deviation term keeps this between-account association distinct from a difficult or interrupted session.

C4: machine-generated against human-authored text. Condition: the classic stories are confirmed to be fixed, human-authored texts, identical for all users within a language. We would predict that the three text components predict reading pace with coefficients of the same sign in generated and in classic texts, and that the interaction of each component with the generated-or-classic indicator is practically equivalent to zero under the registered region. This is a test of exchangeability, and not of quality, and the closest existing evidence is the adult eye-movement corpus on machine-generated text (Bolliger et al., 2025). A non-negligible interaction would mean that the properties governing how children read human-written stories govern their reading of generated ones differently.

Timeline

The export is built and held before submission, so the study can begin on the day in-principle acceptance is confirmed. We anticipate transfer of the extract within four weeks of that confirmation; the frozen audit, the feature extraction with its hand-checked validation, and the outcome-blind dimension reduction within three months of transfer; the confirmatory and sensitivity models within a further four months; and submission of the Stage 2 manuscript within twelve months of in-principle

acceptance. [The author to confirm or adjust these intervals.] Any extension would be negotiated with the handling editor.

Data availability

The session-level extract is held by the data controller and is not in our possession at submission. On in-principle acceptance it will be transferred under a data-sharing agreement between the University of Oxford and the data controller. On acceptance of the Stage 2 manuscript we will deposit, in [the named public repository, with a digital object identifier], the story texts, the computed text features, the component scores, the fitted model objects, the participant-flow counts, the laboratory log, and every table and figure underlying the reported results. Session-level fields that the agreement classes as commercially sensitive may be restricted. For each such field we will state the reason and deposit a derived form sufficient to reproduce every reported analysis, and interested researchers will be able to request the restricted fields through [the named data controller contact or the mechanism the agreement specifies]. The file manifest and the provider-computed aggregate counts contain no record of any individual and will be deposited in the same repository on receipt, before in-principle acceptance. The 35-row structural sample described under Pilot data is deposited there for peer review, under embargo until Stage 2 acceptance, with a view-only link supplied to the editors.

Code availability

All analysis code is version-controlled at https://github.com/pablobernabeu/InTRACt_Interplay_of_Text_Reader_And_Context, in the `preregistration` directory, which will be public and complete at submission and is available for peer review. The theory object with its checks and derived diagram, the design specifications with the precision analysis and its re-solve at 0.95 assurance, the count script supplied to the data controller, and the literature sizing are deposited there, so the theory object, the design analysis and the literature sizing can be reproduced now. The outcome-model

code will be written against simulated data, as described above, and deposited in the same location before the extract is transferred, together with the frozen feature inventory and the archived prior table. The four packages the protocol relies on are cited by their concept identifiers, and the releases in use are theoryforge 0.6.0 (Bernabeu, 2026d), pilotr 0.3.0 (Bernabeu, 2026b), scopusflow 0.4.0 (Bernabeu, 2026c) and depictr 0.3.0 (Bernabeu, 2026a). The session records deposited beside every output record the versions used. The design-analysis figures were composed with depictr, whose audit checks each panel for a floor on rendered type size and for separation in greyscale at the width the journal prints, and refuses to save a panel that fails.

References

Aparicio, X., Mégalakaki, O., Drai-Zerbib, V., & Baccino, T. (2022). Comprehension performances of explanatory texts in French language according to their characteristics: Evidence for 1229 children from 2nd to 9th grade. *Scientific Studies of Reading*, *26*(4), 287–304. https://doi.org/10.1080/10888438.2021.1983819

Balantekin, Y. (2026). Improving students' reading, comprehension, and reading motivation through texts generated from their prompts using ChatGPT-4: A mixed-method study. *Humanities and Social Sciences Communications*, *13*(1), 9. https://doi.org/10.1057/s41599-025-06265-x

Bernabeu, P. (2022). *Language and sensorimotor simulation in conceptual processing: Multilevel analysis and statistical power* [PhD thesis, Lancaster University]. https://doi.org/10.17635/LANCASTER/THESIS/1795

Bernabeu, P. (2026a). *depictr: A unified toolkit for visualising statistical models and data* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21266308

Bernabeu, P. (2026b). *pilotr: Simulate experimental and behavioural data from a portable design*

specification [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21266313

Bernabeu, P. (2026c). *scopusflow: A reproducible workflow layer for Scopus bibliographic searches* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21252669

Bernabeu, P. (2026d). *theoryforge: Systematic theory development* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21229964

Biber, D. (1988). *Variation across speech and writing* (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511621024

Blasi, D. E., Henrich, J., Adamou, E., Kemmerer, D., & Majid, A. (2022). Over-reliance on English hinders cognitive science. *Trends in Cognitive Sciences*, *26*(12), 1153–1170. https://doi.org/10.1016/j.tics.2022.09.015

Bolliger, L. S., Haller, P., Cretton, I. C. R., Reich, D. R., Kew, T., & Jäger, L. A. (2025). EMTeC: A corpus of eye movements on machine-generated texts. *Behavior Research Methods*, *57*(7), 189. https://doi.org/10.3758/s13428-025-02677-4

Booth, J. R., MacWhinney, B., & Harasaki, Y. (2000). Developmental differences in visual and auditory processing of complex sentences. *Child Development*, *71*(4), 981–1003. https://doi.org/10.1111/1467-8624.00203

Booton, S. A., Hodgkiss, A., & Murphy, V. A. (2023). The impact of mobile application features on children's language and literacy learning: A systematic review. *Computer Assisted Language Learning*, *36*(3), 400–429. https://doi.org/10.1080/09588221.2021.1930057

Bradley, V. C., Kuriwaki, S., Isakov, M., Sejdinovic, D., Meng, X.-L., & Flaxman, S. (2021). Unrepresentative big surveys significantly overestimated US vaccine uptake. *Nature*, *600*(7890), 695–700. https://doi.org/10.1038/s41586-021-04198-4

Brossette, B., Persia-Leibnitz, L., Chalbos, M.-J., Prugnières, C., & Ducrot, S. (2026). Measuring

reading time: Comparing logged and self-reported data in relation to reading skills. *PLOS One*, *21*(4), e0344853. https://doi.org/10.1371/journal.pone.0344853

Brysbaert, M., & New, B. (2009). Moving beyond Kučera and Francis: A critical evaluation of current word frequency norms and the introduction of a new and improved word frequency measure for American English. *Behavior Research Methods*, *41*(4), 977–990. https://doi.org/10.3758/BRM.41.4.977

Bulut, A., & Aktepe, V. (2026). The impact of AI-STEM based stories in reading comprehension and motivation. *The Journal of Educational Research*, *119*(4), 469–481. https://doi.org/10.1080/00220671.2026.2633434

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*, 1–28. https://doi.org/10.18637/jss.v080.i01

Cain, K., & Nash, H. M. (2011). The influence of connectives on young readers' processing and comprehension of text. *Journal of Educational Psychology*, *103*(2), 429–441. https://doi.org/10.1037/a0022824

Cain, K., & Oakhill, J. (2014). Reading comprehension and vocabulary: Is vocabulary more important for some aspects of comprehension? *L'Année Psychologique*, *114*(4), 647–662. https://doi.org/10.3917/anpsy.144.0647

Chang, I. (2020). Influences of executive function, language comprehension, and fluency on young children's reading comprehension. *Journal of Early Childhood Research*, *18*(1), 44–57. https://doi.org/10.1177/1476718X19875768

Chatzopoulos, A., Karaflis, A., Kalogiannakis, M., Tzerachoglou, A., Cheirchanteri, G., Sfyroera, E., & Sklavounou, E.-O. (2023). Evaluation of Google Play educational apps for early childhood education. *Advances in Mobile Learning Educational Research*, *3*(2), 770–778. https://doi.org/

10.25082/AMLER.2023.02.004

Clark, H. H. (1973). The language-as-fixed-effect fallacy: A critique of language statistics in psychological research. *Journal of Verbal Learning and Verbal Behavior*, *12*(4), 335–359. https://doi.org/10.1016/s0022-5371(73)80014-3

Clinton-Lisell, V. (2022). Listening ears or reading eyes: A meta-analysis of reading and listening comprehension comparisons. *Review of Educational Research*, *92*(4), 543–582. https://doi.org/10.3102/00346543211060871

Clinton-Lisell, V. (2023). Does reading while listening to text improve comprehension compared to reading only? A systematic review and meta-analysis. *Educational Research: Theory and Practice*, *34*(3), 133–155. https://eric.ed.gov/?id=EJ1403866

Crossley, S. A. (2025). Developing linguistic constructs of text readability using natural language processing. *Scientific Studies of Reading*, *29*(2), 138–160. https://doi.org/10.1080/10888438.2024.2422365

Crossley, S. A., Kyle, K., & Dascalu, M. (2019). The Tool for the Automatic Analysis of Cohesion 2.0: Integrating semantic similarity and text overlap. *Behavior Research Methods*, *51*(1), 14–27. https://doi.org/10.3758/s13428-018-1142-4

Crossley, S. A., Skalicky, S., Berger, C., & Heidari, A. (2023). Assessing readability formulas in the wild. In M. Dascalu, P. Marti, & F. Pozzi (Eds.), *Polyphonic Construction of Smart Learning Ecosystems* (pp. 91–101). Springer Nature. https://doi.org/10.1007/978-981-19-5240-1_6

Cui, N., Wang, Y., Luo, J., & Wu, Y. (2024). The role of executive functions in 9- to 12-year-old children's sentence processing: An eye-movement study. *Journal of Research in Reading*, *47*(2), 201–219. https://doi.org/10.1111/1467-9817.12449

Delage, H., & Frauenfelder, U. H. (2019). Syntax and working memory in typically-developing

children: Focus on syntactic complexity. *Language, Interaction and Acquisition*, *10*(2), 141–176. https://doi.org/10.1075/lia.18013.del

Dhami, M. K., Hertwig, R., & Hoffrage, U. (2004). The role of representative design in an ecological approach to cognition. *Psychological Bulletin*, *130*(6), 959–988. https://doi.org/10.1037/0033-2909.130.6.959

Faust, M. E., Balota, D. A., Spieler, D. H., & Ferraro, F. R. (1999). Individual differences in information-processing rate and amount: Implications for group differences in response latency. *Psychological Bulletin*, *125*(6), 777–799. https://doi.org/10.1037/0033-2909.125.6.777

Fisher, A. J., Medaglia, J. D., & Jeronimus, B. F. (2018). Lack of group-to-individual generalizability is a threat to human subjects research. *Proceedings of the National Academy of Sciences*, *115*(27), E6106–E6115. https://doi.org/10.1073/pnas.1711978115

Freed, J., & Cain, K. (2021). Assessment of inference-making in children using comprehension questions and story retelling: Effect of text modality and a story presentation format. *International Journal of Language & Communication Disorders*, *56*(3), 637–652. https://doi.org/10.1111/1460-6984.12620

Furlong, L., Serry, T., Erickson, S., Lefort, M., Gath, M., & Morris, M. (2025). Finding the needle in the haystack: A quality appraisal of mobile applications for foundational literacy skills. *Early Childhood Education Journal*. https://doi.org/10.1007/s10643-025-02066-8

Gibson, E. (1998). Linguistic complexity: Locality of syntactic dependencies. *Cognition*, *68*(1), 1–76. https://doi.org/10.1016/S0010-0277(98)00034-1

Giusto, M., & Ehri, L. C. (2019). Effectiveness of a partial read-aloud test accommodation to assess reading comprehension in students with a reading disability. *Journal of Learning Disabilities*, *52*(3), 259–270. https://doi.org/10.1177/0022219418789377

Goldman, S. R. (2024). Reading is complex: Implications for research and practice. *Educational Psychologist*, *59*(4), 315–325. https://doi.org/10.1080/00461520.2024.2418062

Goldstone, R. L., & Lupyan, G. (2016). Discovering psychological principles by mining naturally occurring data sets. *Topics in Cognitive Science*, *8*(3), 548–568. https://doi.org/10.1111/tops.12212

Graesser, A. C., McNamara, D. S., Cai, Z., Conley, M., Li, H., & Pennebaker, J. (2014). Coh-metrix measures text characteristics at multiple levels of language and discourse. *The Elementary School Journal*, *115*(2), 210–229. https://doi.org/10.1086/678293

Grosz, M. P., Rohrer, J. M., & Thoemmes, F. (2020). The taboo against explicit causal inference in nonexperimental psychology. *Perspectives on Psychological Science*, *15*(5), 1243–1255. https://doi.org/10.1177/1745691620921521

Helder, A., Van Leijenhorst, L., & Van den Broek, P. (2016). Coherence monitoring by good and poor comprehenders in elementary school: Comparing offline and online measures. *Learning and Individual Differences*, *48*, 17–23. https://doi.org/10.1016/j.lindif.2016.02.008

Hernán, M. A. (2018). The C-word: Scientific euphemisms do not improve causal inference from observational data. *American Journal of Public Health*, *108*(5), 616–619. https://doi.org/10.2105/ajph.2018.304337

Hofman, J. M., Sharma, A., & Watts, D. J. (2017). Prediction and explanation in social systems. *Science*, *355*(6324), 486–488. https://doi.org/10.1126/science.aal3856

Hofman, J. M., Watts, D. J., Athey, S., Garip, F., Griffiths, T. L., Kleinberg, J., Margetts, H., Mullainathan, S., Salganik, M. J., Vazire, S., Vespignani, A., & Yarkoni, T. (2021). Integrating explanation and prediction in computational social science. *Nature*, *595*(7866), 181–188. https://doi.org/10.1038/s41586-021-03659-0

Holleman, G. A., Hooge, I. T. C., Kemner, C., & Hessels, R. S. (2020). The "real-world approach" and its problems: A critique of the term ecological validity. *Frontiers in Psychology*, *11*, 721. https://doi.org/10.3389/fpsyg.2020.00721

Horn, J. L. (1965). A rationale and test for the number of factors in factor analysis. *Psychometrika*, *30*(2), 179–185. https://doi.org/10.1007/BF02289447

Hui, B., & Godfroid, A. (2026). Listening, reading, or both? Rethinking the comprehension benefits of reading-while-listening. *Language Learning*, *76*(1), 311–351. https://doi.org/10.1111/lang.12721

Joseph, H. S. S. L., & Liversedge, S. P. (2013). Children's and adults' on-line processing of syntactically ambiguous sentences during reading. *PLoS ONE*, *8*(1), e54141. https://doi.org/10.1371/journal.pone.0054141

Joseph, H. S. S. L., Nation, K., & Liversedge, S. P. (2013). Using Eye Movements to Investigate Word Frequency Effects in Children's Sentence Reading. *School Psychology Review*, *42*(2), 207–222. https://doi.org/10.1080/02796015.2013.12087485

Judd, C. M., Westfall, J., & Kenny, D. A. (2012). Treating stimuli as a random factor in social psychology: A new and comprehensive solution to a pervasive but largely ignored problem. *Journal of Personality and Social Psychology*, *103*(1), 54–69. https://doi.org/10.1037/a0028347

Kaup, B., Günther, F., & Dudschig, C. (2025). The interplay of modal and amodal processes in language comprehension at the word level and beyond. In *Psychology of Learning and Motivation*. Academic Press. https://doi.org/10.1016/bs.plm.2025.08.001

Keelor, J. L., Creaghead, N. A., Silbert, N. H., Breit, A. D., & Horowitz-Kraus, T. (2023). Impact of text-to-speech features on the reading comprehension of children with reading and language difficulties. *Annals of Dyslexia*, *73*(3), 469–486. https://doi.org/10.1007/s11881-023-00281-9

Kihlstrom, J. F. (2021). Ecological validity and "ecological validity." *Perspectives on Psychological Science*, *16*(2), 466–471. https://doi.org/10.1177/1745691620966791

Kleijn, S., Pander Maat, H. L. W., & Sanders, T. J. M. (2019). Comprehension effects of connectives across texts, readers, and coherence relations. *Discourse Processes*, *56*(5-6), 447–464. https://doi.org/10.1080/0163853x.2019.1605257

Korochkina, M., Marelli, M., Brysbaert, M., & Rastle, K. (2024). The Children and Young People's Books Lexicon (CYP-LEX): A large-scale lexical database of books read by children and young people in the United Kingdom. *Quarterly Journal of Experimental Psychology*, *77*(12), 2418–2438. https://doi.org/10.1177/17470218241229694

Kuperman, V., Buzato, D., & Rothe-Neves, R. (2025). Global measures of syntactic and lexical complexity are not strong predictors of eye-movement patterns in sentence and passage reading. *Quarterly Journal of Experimental Psychology*, *78*(8), 1675–1690. https://doi.org/10.1177/17470218251317372

Kuperman, V., Schroeder, S., & Gnetov, D. (2024). Word length and frequency effects on text reading are highly similar in 12 alphabetic languages. *Journal of Memory and Language*, *135*, 104497. https://doi.org/10.1016/j.jml.2023.104497

Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The parable of Google Flu: Traps in big data analysis. *Science*, *343*(6176), 1203–1205. https://doi.org/10.1126/science.1248506

Ludewig, U., Hübner, N., & Schroeder, S. (2023). Vocabulary, text coverage, word frequency and the lexical threshold in elementary school reading comprehension. *Reading and Writing*, *36*(9), 2409–2431. https://doi.org/10.1007/s11145-022-10385-0

MacDonald, M. C., & Christiansen, M. H. (2002). Reassessing working memory: Comment on Just and Carpenter (1992) and Waters and Caplan (1996). *Psychological Review*, *109*(1), 35–54.

https://doi.org/10.1037/0033-295x.109.1.35

Magnabosco, F., & Hauk, O. (2024). An eye on semantics: A study on the influence of concreteness and predictability on early fixation durations. *Language, Cognition and Neuroscience*, *39*(3), 302–316. https://doi.org/10.1080/23273798.2023.2274558

Marjokorpi, J., & van Rijt, J. (2025). Grammatical understanding predicts reading comprehension in secondary-level students: Insights from a Finnish national survey. *Language and Education*, *39*(4), 924–943. https://doi.org/10.1080/09500782.2024.2376280

Martinc, M., Pollak, S., & Robnik-Šikonja, M. (2021). Supervised and unsupervised neural approaches to text readability. *Computational Linguistics*, *47*(1), 141–179. https://doi.org/10.1162/coli_a_00398

McCarthy, P. M., & Jarvis, S. (2010). MTLD, vocd-D, and HD-D: A validation study of sophisticated approaches to lexical diversity assessment. *Behavior Research Methods*, *42*(2), 381–392. https://doi.org/10.3758/BRM.42.2.381

McNamara, D. S., Kintsch, E., Songer, N. B., & Kintsch, W. (1996). Are Good Texts Always Better? Interactions of Text Coherence, Background Knowledge, and Levels of Understanding in Learning From Text. *Cognition and Instruction*, *14*(1), 1–43. https://doi.org/10.1207/s1532690xci1401_1

McNamara, D. S., Ozuru, Y., & Floyd, R. G. (2011). Comprehension challenges in the fourth grade: The roles of text cohesion, text genre, and readers' prior knowledge. *International Electronic Journal of Elementary Education*, *4*(1), 229–257. https://files.eric.ed.gov/fulltext/EJ1070456.pdf

Meng, X.-L. (2018). Statistical paradises and paradoxes in big data (I): Law of large populations, big data paradox, and the 2016 US presidential election. *The Annals of Applied Statistics*, *12*(2),

685–726. https://doi.org/10.1214/18-aoas1161sf

Montag, J. L., & MacDonald, M. C. (2015). Text exposure predicts spoken production of complex sentences in 8- and 12-year-old children and adults. *Journal of Experimental Psychology: General*, *144*(2), 447–468. https://doi.org/10.1037/xge0000054

Muraki, E. J., Reggin, L. D., Feddema, C. Y., & Pexman, P. M. (2025). The development of abstract word meanings. *Journal of Child Language*, *52*(1), 195–207. https://doi.org/10.1017/S0305000923000569

Nastase, S. A., Goldstein, A., & Hasson, U. (2020). Keep it real: Rethinking the primacy of experimental control in cognitive neuroscience. *NeuroImage*, *222*, 117254. https://doi.org/10.1016/j.neuroimage.2020.117254

Nie, Y., Deacon, H., Fyshe, A., & Epp, C. D. (2022). Predicting reading comprehension scores of elementary school students. In A. Mitrovic & N. Bosch (Eds.), *Proceedings of the 15th International Conference on Educational Data Mining* (pp. 158–170). International Educational Data Mining Society. https://doi.org/10.5281/ZENODO.6852952

Nielsen, J. L., Christensen, R. V., & Poulsen, M. (2025). Syntactic comprehension—a separate source of individual variance in middle-school children's reading comprehension. *Reading Research Quarterly*, *60*(2), e70003. https://doi.org/10.1002/rrq.70003

O'Reilly, T., & McNamara, D. S. (2007). Reversing the reverse cohesion effect: Good texts can be better for strategic, high-knowledge readers. *Discourse Processes*, *43*(2), 121–152. https://doi.org/10.1080/01638530709336895

Paivio, A. (1991). Dual coding theory: Retrospect and current status. *Canadian Journal of Psychology / Revue Canadienne de Psychologie*, *45*(3), 255–287. https://doi.org/10.1037/h0084295

Parry, D. A., Davidson, B. I., Sewall, C. J. R., Fisher, J. T., Mieczkowski, H., & Quintana, D. S.

(2021). A systematic review and meta-analysis of discrepancies between logged and self-reported digital media use. *Nature Human Behaviour*, *5*(11), 1535–1547. https://doi.org/10.1038/s41562-021-01117-5

Paxton, A., & Griffiths, T. L. (2017). Finding the traces of behavioral and cognitive processes in big data and naturally occurring datasets. *Behavior Research Methods*, *49*(5), 1630–1638. https://doi.org/10.3758/s13428-017-0874-x

Peng, P., Barnes, M., Wang, C., Wang, W., Li, S., Swanson, H. L., Dardick, W., & Tao, S. (2018). A meta-analysis on the relation between reading and working memory. *Psychological Bulletin*, *144*(1), 48–76. https://doi.org/10.1037/bul0000124

Pickren, S. E., Stacy, M., Tufo, S. N. D., Spencer, M., & Cutting, L. E. (2022). The contribution of text characteristics to reading comprehension: Investigating the influence of text emotionality. *Reading Research Quarterly*, *57*(2), 649–667. https://doi.org/10.1002/rrq.431

Ratner, S., Ong, C., Mathers, S., & Murphy, V. A. (2025). "Did a robot write that?": AI-generated digital storybooks. *ELT Journal*, *79*(4), 595–604. https://doi.org/10.1093/elt/ccaf034

Rayner, K., Schotter, E. R., Masson, M. E. J., Potter, M. C., & Treiman, R. (2016). So much to read, so little time: How do we read, and can speed reading help? *Psychological Science in the Public Interest*, *17*(1), 4–34. https://doi.org/10.1177/1529100615623267

Revelle, W. (2023). *Psych: Procedures for psychological, psychometric, and personality research* [Manual]. Northwestern University. https://doi.org/10.32614/CRAN.package.psych

Ripoll Salceda, J. C., Tapia Montesinos, M. M., & Aguado Alonso, G. (2020). Velocidad lectora en alumnado hispanohablante: Un metaanálisis. *Revista de Psicodidáctica*, *25*(2), 158–165. https://doi.org/10.1016/j.psicod.2020.01.002

Rohrer, J. M. (2018). Thinking clearly about correlations and causation: Graphical causal models

for observational data. *Advances in Methods and Practices in Psychological Science*, *1*(1), 27–42. https://doi.org/10.1177/2515245917745629

Salganik, M. J., Lundberg, I., Kindel, A. T., Ahearn, C. E., Al-Ghoneim, K., Almaatouq, A., Altschul, D. M., Brand, J. E., Carnegie, N. B., Compton, R. J., Datta, D., Davidson, T., Filippova, A., Gilroy, C., Goode, B. J., Jahani, E., Kashyap, R., Kirchner, A., McKay, S., … McLanahan, S. (2020). Measuring the predictability of life outcomes with a scientific mass collaboration. *Proceedings of the National Academy of Sciences*, *117*(15), 8398–8403. https://doi.org/10.1073/pnas.1915006117

Sen, I., Flöck, F., Weller, K., Weiß, B., & Wagner, C. (2021). A total error framework for digital traces of human behavior on online platforms. *Public Opinion Quarterly*, *85*(S1), 399–422. https://doi.org/10.1093/poq/nfab018

Siegelman, N., Schroeder, S., Acartürk, C., Ahn, H.-D., Alexeeva, S., Amenta, S., Bertram, R., Bonandrini, R., Brysbaert, M., Chernova, D., Da Fonseca, S. M., Dirix, N., Duyck, W., Fella, A., Frost, R., Gattei, C. A., Kalaitzi, A., Kwon, N., Lõo, K., … Kuperman, V. (2022). Expanding horizons of cross-linguistic research on reading: The Multilingual Eye-movement Corpus (MECO). *Behavior Research Methods*, *54*(6), 2843–2863. https://doi.org/10.3758/s13428-021-01772-6

Siegelman, N., Schroeder, S., Bao, Y. B., Acartürk, C., Agrawal, N., Bolliger, L. S., Brasser, J., Campos-Rojas, C., Drieghe, D., Filipović Đurđević, D., Goldina, S., Ibáñez Orellana, R., Jäger, L. A., Jóhannesson, Ó. I., Khare, A., Kharlamov, N., Knudsen, H. B. S., Kristjánsson, Á., Lee, C. E., … Kuperman, V. (2025). Wave 2 of the Multilingual Eye-Movement Corpus (MECO): New text reading data across languages. *Scientific Data*, *12*(1), 1183. https://doi.org/10.1038/s41597-025-05453-3

Sortkær, B., Smith, E., Reimer, D., Oehmcke, S., & Andersen, I. G. (2021). Both sides of the story: Comparing student-level data on reading performance from administrative registers to application generated data from a reading app. *EPJ Data Science*, *10*(1), 44. https://doi.org/10.1140/epjds/s13688-021-00300-y

Torppa, M., Niemi, P., Vasalampi, K., Lerkkanen, M.-K., Tolvanen, A., & Poikkeus, A.-M. (2020). Leisure reading (but not any kind) and reading comprehension support each other—a longitudinal study across Grades 1 and 9. *Child Development*, *91*(3), 876–900. https://doi.org/10.1111/cdev.13241

Trauzettel-Klosinski, S., & Dietz, K. (2012). Standardized assessment of reading performance: The New International Reading Speed Texts IReST. *Investigative Ophthalmology & Visual Science*, *53*(9), 5452–5461. https://doi.org/10.1167/iovs.11-8284

Valentini, A., Pye, R. E., Houston-Price, C., Ricketts, J., & Kirkby, J. A. (2024). Online processing shows advantages of bimodal listening-while-reading for vocabulary learning: An eye-tracking study. *Reading Research Quarterly*, *59*(1), 79–101. https://doi.org/10.1002/rrq.522

van Bergen, E., Vasalampi, K., & Torppa, M. (2021). How are practice and performance related? Development of reading from age 5 to 15. *Reading Research Quarterly*, *56*(3), 415–434. https://doi.org/10.1002/rrq.309

Van den Akker, O. R., Weston, S., Campbell, L., Chopik, B., Damian, R., Davis-Kean, P., Hall, A., Kosie, J., Kruse, E., Olsen, J., Ritchie, S., Valentine, K., Van 't Veer, A., & Bakker, M. (2021). Preregistration of secondary data analysis: A template and tutorial. *Meta-Psychology*, *5*. https://doi.org/10.15626/mp.2020.2625

van Heuven, W. J. B., Mandera, P., Keuleers, E., & Brysbaert, M. (2014). Subtlex-UK: A new and improved word frequency database for British English. *Quarterly Journal of Experimental*

Psychology, *67*(6), 1176–1190. https://doi.org/10.1080/17470218.2013.850521

Wasito Kirana, A., Segoh, D., Achmad Syukri, M., & Prassetyo, S. (2025). Sentence processing in Indonesian-English bilingual children with and without language impairment. *GEMA Online® Journal of Language Studies*, *25*(4), 796–816. https://doi.org/10.17576/gema-2025-2504-01

Weston, S. J., Ritchie, S. J., Rohrer, J. M., & Przybylski, A. K. (2019). Recommendations for increasing the transparency of analysis of preexisting data sets. *Advances in Methods and Practices in Psychological Science*, *2*(3), 214–227. https://doi.org/10.1177/2515245919848684

Wonnacott, E., Joseph, H. S. S. L., Adelman, J. S., & Nation, K. (2016). Is children's reading "good enough"? Links between online processing and comprehension as children read syntactically ambiguous sentences. *Quarterly Journal of Experimental Psychology*, *69*(5), 855–879. https://doi.org/10.1080/17470218.2015.1011176

Wood, S. G., Moxley, J. H., Tighe, E. L., & Wagner, R. K. (2018). Does use of text-to-speech and related read-aloud tools improve reading comprehension for students with reading disabilities? A meta-analysis. *Journal of Learning Disabilities*, *51*(1), 73–84. https://doi.org/10.1177/0022219416688170

Yarkoni, T. (2022). The generalizability crisis. *Behavioral and Brain Sciences*, *45*, e1. https://doi.org/10.1017/s0140525x20001685

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Competing interests

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Figures

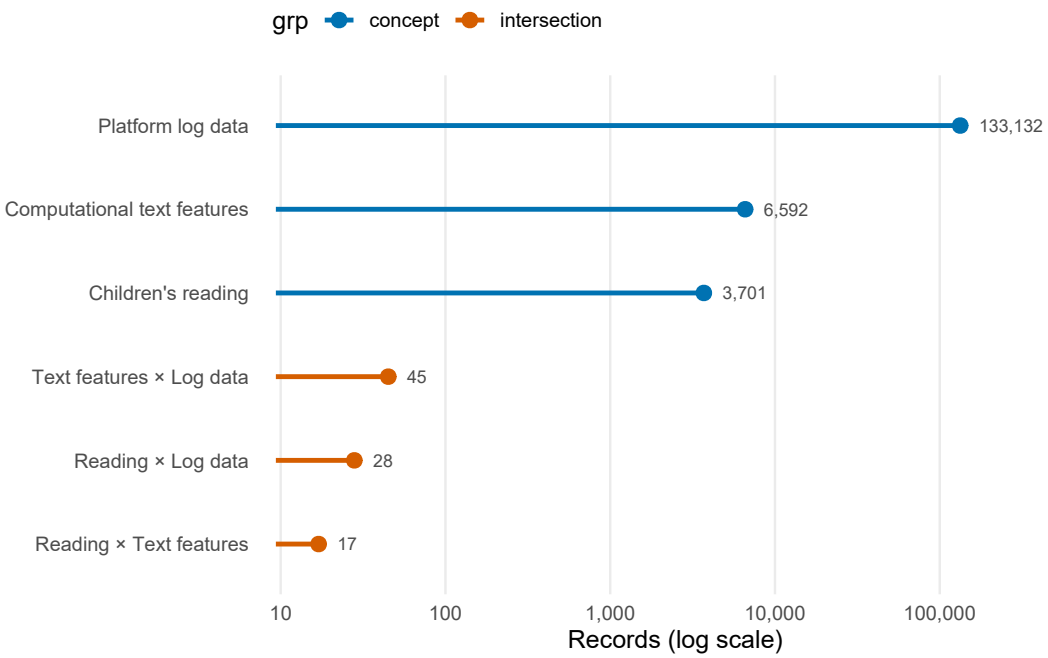


Figure 1

Sizing the literature around this study. Records on Scopus for 2018 to 2026, for children's reading comprehension, computational text features and platform log data, and for every intersection of the three. The pairwise intersections hold 17, 28 and 45 records, and the three-way intersection holds none, so it does not appear on the logarithmic axis. Search expressions and the date of the search are in the supplementary record.

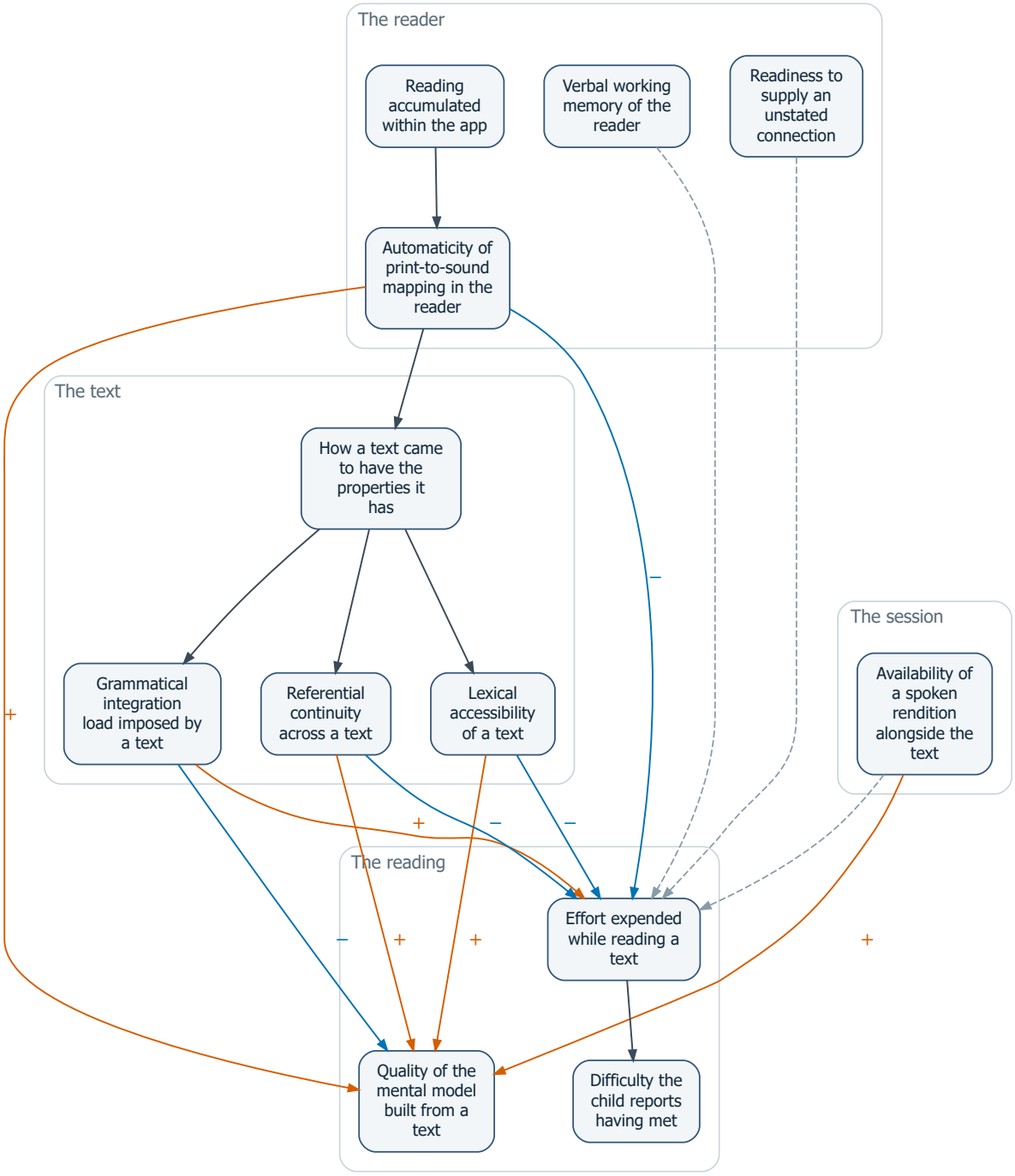


Figure 2

The nomological net, generated from the deposited theory object. Solid arrows are causal paths, with the predicted sign shown on the line where the theory fixes one, and dashed lines mark moderation. Text properties and reader capacities converge on the effort a reading costs, which shapes the difficulty the child reports. A parallel set of paths reaches the quality of the mental model built, which these data can measure only if comprehension items prove to exist. How a text came to have its properties, meaning the generation route and the reading level being served, is a common cause of all three text properties and receives a path from the reader's own decoding automaticity, which is why every text coefficient is estimated within reading level.

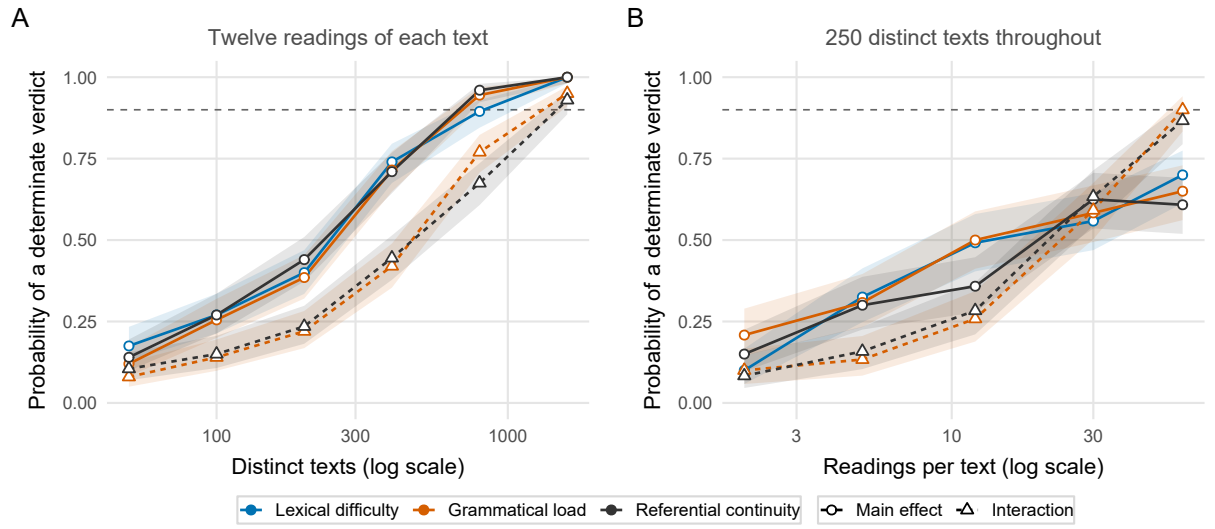


Figure 3

Precision as a function of what there is more of. Panel a sweeps the number of distinct texts at twelve readings each, and panel b holds the number of texts at 250 and sweeps readings per text. Points are the probability that a 95 per cent interval falls determinately outside the region of practical equivalence for an effect of 0.10 standard deviations, over 200 simulation replicates per grid point in a and 120 in b; shaded bands are 95 per cent Wilson score intervals on each probability, and the dashed line marks the 0.90 target. The two panels answer different questions, and only one of them is about having more data of the kind this platform generates most readily.